6 Surface area and volume

LESSON SEQUENCE

6.1 Overview 410
6.2 Area 414
6.3 Total surface area 424
6.4 Volume 435
6.5 Review 449



LESSON
6.1 Overview

Why learn this?



People must measure! How much extra freezer space is required for the amount of ice cream on your birthday party? How many litres of water will it take to fill the new pool? These are just a few examples where measurement skills are needed.

Measuring tools have advanced significantly in their capability to measure extremely small and extremely large amounts and objects, leading to many breakthroughs in medicine, engineering, science, architecture and astronomy.

In medical research, measurement errors can occur when measuring factors such as blood pressure, heart rate, or hormone levels. This can lead to incorrect conclusions about the effectiveness of the medication. In environmental research, measurement errors can occur when measuring factors such as air pollution, water quality or temperature. For example, a study investigating the impact of air pollution on respiratory health may yield biased results if the air pollution measurements are inaccurate due to measurement error. This can lead to incorrect conclusions about the impact of air pollution on respiratory health.

Have you ever wondered why tennis balls are sold in cylindrical containers? This is an example of manufacturers wanting to minimise the amount of waste in packaging. Measurement data impacted by error can result in biased findings because the data does not accurately represent the true values of the factors being measured. This can lead to incorrect conclusions about the relationships between factors or the effectiveness of interventions.

Hey students! Bring these pages to life online

Watch videos

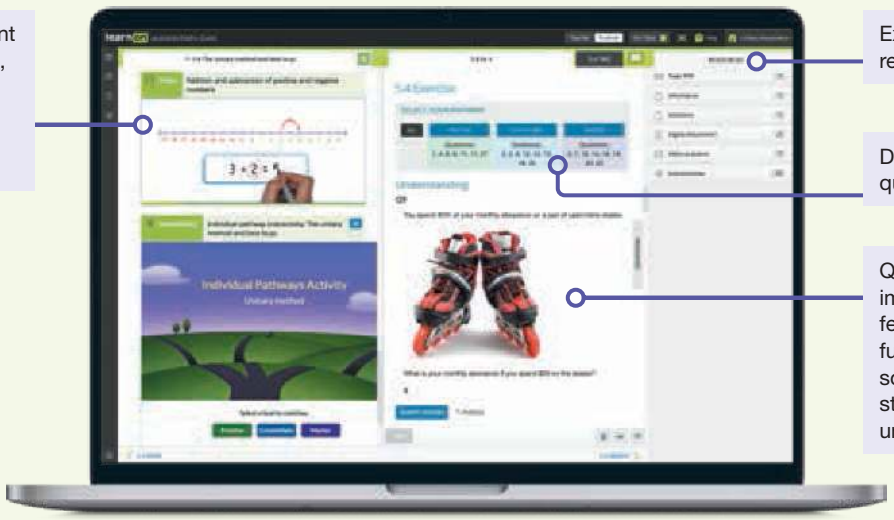
Engage with interactivities

Answer questions and check solutions

Find all this and MORE in jacPLUS



Reading content and rich media, including interactivities and videos for every concept



Extra learning resources

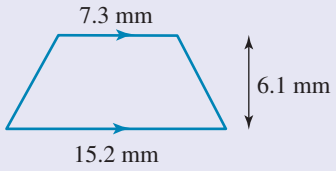
Differentiated question sets

Questions with immediate feedback, and fully worked solutions to help students get unstuck

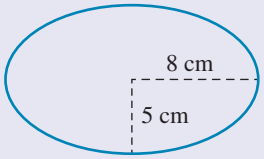
Exercise 6.1 Pre-test

learnon

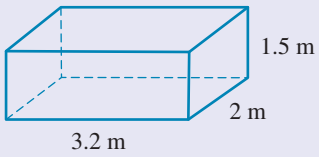
1. Calculate the area of the shape, correct to 2 decimal places.



2. Calculate the area of the ellipse, correct to 1 decimal place.

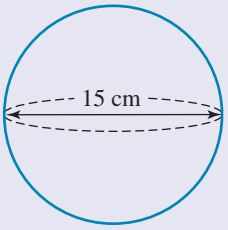


3. MC Select the total surface area of the rectangular prism from the following.

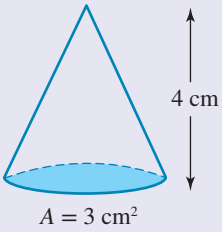


A. $9.6\,m^2$ **B.** $14.2\,m^2$ **C.** $22.0\,m^2$ **D.** $28.4\,m^2$

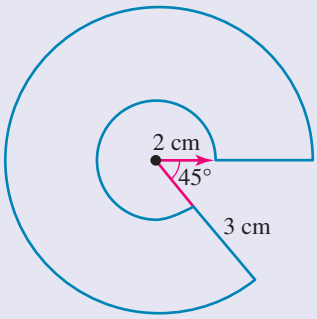
4. Calculate the total surface area of the sphere, correct to 1 decimal place.



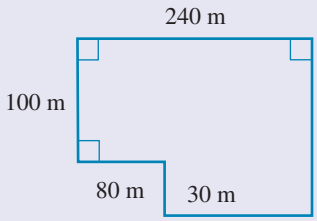
5. Calculate the volume of the solid.



6. Calculate the area of the shape, correct to 1 decimal place.

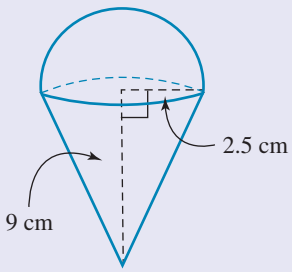


7. A council park is shown below.



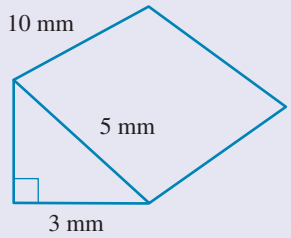
A worker charges \$30 per 1000 m^2 to mow the grass. Determine how much it will cost the council to have the grass mown.

8. MC Select the total surface area of the object shown from the following.

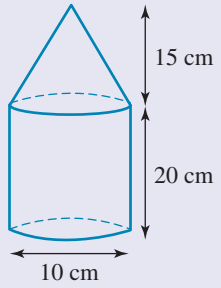


A. 109.96 cm^2 **B.** 112.63 cm^2 **C.** 151.9 cm^2 **D.** 124.36 cm^2

9. Determine the volume of the triangular prism.

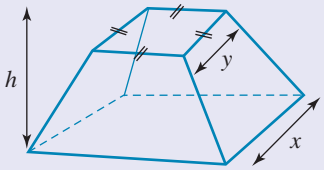


10. MC Select the volume of the object from the following.



A. $2748.9\,\text{cm}^3$ **B.** $1701.7\,\text{cm}^3$ **C.** $1963.5\,\text{cm}^3$ **D.** $7854\,\text{cm}^3$

11. MC The volume of the frustum of a square-based pyramid is given by:



A. $V = \frac{1}{3}\pi\ (x^2 + y^2)$ **B.** $V = \frac{1}{3}\pi\ (x^2 + xy + y^2)$

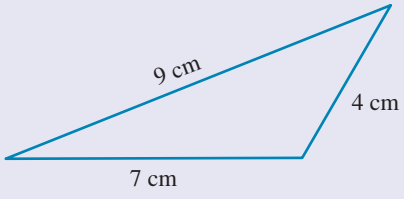
C. $V = \frac{1}{3}h\pi\ (x^2 + xy + y^2)$ **D.** $V = \frac{1}{3}h\ (x^2 + xy + y^2)$

12. The volume of a ball is given by the formula $V = \frac{4}{3}\pi r^3$. Evaluate the radius of a ball with a volume of $384.66\,\text{cm}^3$. Give your answer correct to 1 decimal place.

13. MC Determine what effect doubling the radius and halving the height of a cone will have on its volume.

A. The volume will be the same.
B. The volume will be halved.
C. The volume will be doubled.
D. The volume will be quadrupled.

14. Using Heron's formula, evaluate the area of the triangle, correct to 1 decimal place.



15. A cylindrical soft drink can has a diameter of 6.4 cm and a height of 14.3 cm. If the can is only half full, determine what capacity of soft drink remains, to the nearest millilitre.

LESSON
6.2 Area

LEARNING INTENTIONS

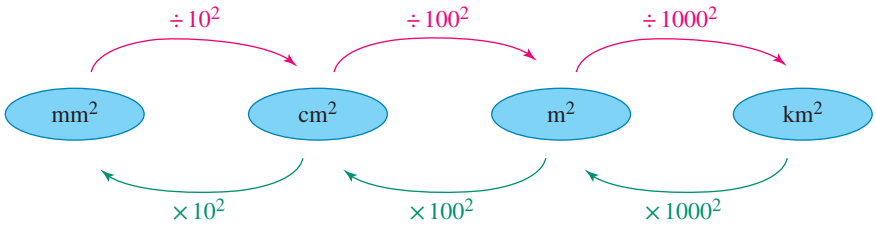
At the end of this lesson you should be able to:

- convert between units of area
- calculate the areas of plane figures using area formulas
- calculate the area of a triangle using Heron's formula.

6.2.1 Area

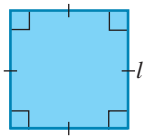
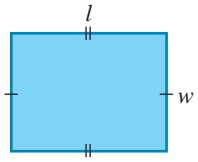
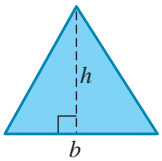
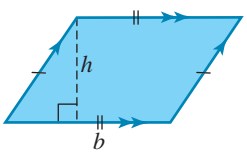
eles-4809

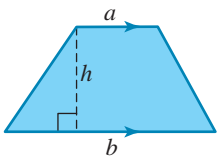
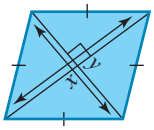
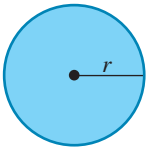
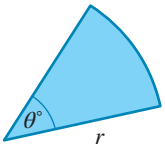
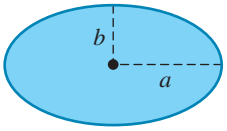
- The **area** of a figure is the amount of surface covered by the figure.
- The units used for area are mm^2, cm^2, m^2, km^2 and ha (hectares).
- One unit that is often used when measuring land is the hectare. It is equal to $10\,000\ m^2$.
- The following diagram can be used to convert between units of area.



Area formulas

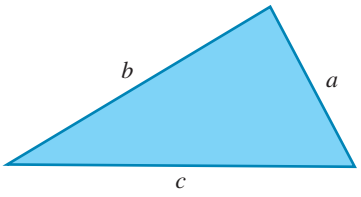
- The table below shows the formulas for the area of some common shapes.

Shape	Diagram	Formula
Square		$A = l^2$
Rectangle		$A = lw$
Triangle		$A = \frac{1}{2}bh$
Parallelogram		$A = bh$

Shape	Diagram	Formula
Trapezium		$A = \frac{1}{2}(a+b)h$
Kite (including rhombus)		$A = \frac{1}{2}xy$
Circle		$A = \pi r^2$
Sector		$A = \frac{\theta°}{360°} \times \pi r^2$
Ellipse		$A = \pi ab$

Heron's formula

- The area of a triangle can be calculated if the lengths of all three sides are known.



- The area, A, of a triangle given the lengths of the three sides a, b and c is:

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \frac{a+b+c}{2}$, the semi-perimeter.

Digital technology

When using the number π in calculations, it is best to use a calculator. Calculators use the exact value of π, which will ensure your answer is exactly correct.

The π button on the TI-*n*spire CX CAS calculator is found near the bottom left of the calculator, as can be seen in the image.

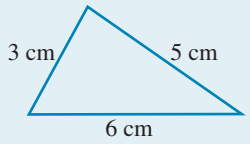


If you do not have a calculator to hand, you can use the approximations $\pi \approx \frac{22}{7} \approx 3.14$; however, your answer may differ from the exact answer by a small amount.

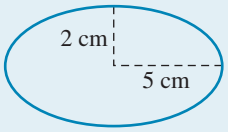
WORKED EXAMPLE 1 Calculating areas of plane figures

Calculate the areas of the following plane figures, correct to 2 decimal places.

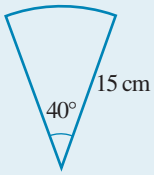
a.



b.



c.



THINK	WRITE
a. 1. Three side lengths are known; apply Heron's formula.	**a.** $A = \sqrt{s(s-a)(s-b)(s-c)}$
2. Identify the values of a, b and c.	$a = 3, b = 5, c = 6$
3. Calculate the value of s, the semi-perimeter of the triangle.	$s = \frac{a+b+c}{2}$ $= \frac{3+5+6}{2}$ $= \frac{14}{2}$ $= 7$
4. Substitute the values of a, b, c and s into Heron's formula and evaluate, correct to 2 decimal places.	$A = \sqrt{7(7-3)(7-5)(7-6)}$ $= \sqrt{7 \times 4 \times 2 \times 1}$ $= \sqrt{56}$ $= 7.48 \text{ cm}^2$
b. 1. The shape shown is an ellipse. Write the appropriate area formula.	**b.** $A = \pi ab$
2. Identify the values of a and b (the semi-major and semi-minor axes).	$a = 5, b = 2$

3. Substitute the values of a and b into the formula and evaluate, correct to 2 decimal places. ($\pi = 3.14$)

$$A = \pi \times 5 \times 2$$
$$= 31.42\,\text{cm}^2$$

c. 1. The shape shown is a sector. Write the formula for finding the area of a sector.

c. $$A = \frac{\theta}{360°} \times \pi r^2$$

2. Write the value of θ and r.

$$\theta = 40°,\ r = 15$$

3. Substitute and evaluate the expression, correct to 2 decimal places.

$$A = \frac{40°}{360°} \times \pi \times 15^2$$
$$= 78.54\,\text{cm}^2$$

6.2.2 Areas of composite figures

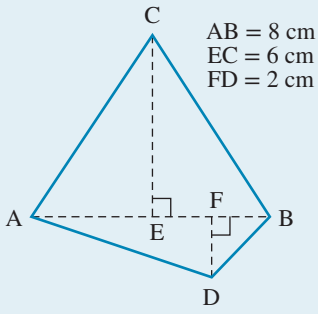
eles-4810

- A **composite figure** is a figure made up of a combination of simple figures.
- The area of a composite figure can be calculated by:
  - calculating the sum of the areas of the simple figures that make up the composite figure
  - calculating the area of a larger shape and then subtracting the extra area involved.

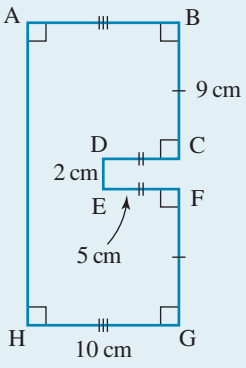
WORKED EXAMPLE 2 Calculating areas of composite shapes

Calculate the area of each of the following composite shapes.

a.



b.



THINK

a. 1. ACBD is a quadrilateral that can be split into two triangles: ΔABC and ΔABD.

2. Write the formula for the area of a triangle containing base and height.

3. Identify the values of b and h for ΔABC.

4. Substitute the values of the pronumerals into the formula and calculate the area of ΔABC.

5. Identify the values of b and h for ΔABD.

WRITE

a. Area ACBD = Area ΔABC + Area ΔABD

$$A_{\text{triangle}} = \frac{1}{2}bh$$

ΔABC: $b = \text{AB} = 8,\ h = \text{EC} = 6$

$$\text{Area of } \Delta\text{ABC} = \frac{1}{2} \times \text{AB} \times \text{EC}$$
$$= \frac{1}{2} \times 8 \times 6$$
$$= 24\,\text{cm}^2$$

ΔABD: $b = \text{AB} = 8, h = \text{FD} = 2$

6. Calculate the area of ΔABD.

$$\text{Area of } \Delta ABD = \frac{1}{2} AB \times FD$$
$$= \frac{1}{2} \times 8 \times 2$$
$$= 8 \text{ cm}^2$$

7. Add the areas of the two triangles together to find the area of the quadrilateral ACBD.

$$\text{Area of } ACBD = 24 \text{ cm}^2 + 8 \text{ cm}^2$$
$$= 32 \text{ cm}^2$$

b. 1. One way to calculate the area of the shape shown is to find the total area of the rectangle ABGH and then subtract the area of the smaller rectangle DEFC.

b. Area = Area ABGH − Area DEFC

2. Write the formula for the area of a rectangle.

$$A_{\text{rectangle}} = l \times w$$

3. Identify the values of the pronumerals for the rectangle ABGH.

Rectangle ABGH: $l = 9 + 2 + 9$
$= 20$
$w = 10$

4. Substitute the values of the pronumerals into the formula to find the area of the rectangle ABGH.

$$\text{Area of ABGH} = 20 \times 10$$
$$= 200 \text{ cm}^2$$

5. Identify the values of the pronumerals for the rectangle DEFC.

Rectangle DEFC: $l = 5$, $w = 2$

6. Substitute the values of the pronumerals into the formula to find the area of the rectangle DEFC.

$$\text{Area of DEFC} = 5 \times 2$$
$$= 10 \text{ cm}^2$$

7. Subtract the area of the rectangle DEFC from the area of the rectangle ABGH to find the area of the given shape.

$$\text{Area} = 200 - 10$$
$$= 190 \text{ cm}^2$$

on Resources

eWorkbook Topic 6 workbook (worksheets, code puzzle and a project) (ewbk-13290)

Video eLesson Composite area (eles-1886)

Interactivities Individual pathway interactivity: Area (int-4593)
Conversion chart for area (int-3783)
Area of rectangles (int-3784)
Area of parallelograms (int-3786)
Area of trapeziums (int-3790)
Area of a kite (int-6136)
Area of circles (int-3788)
Area of a sector (int-6076)
Area of an ellipse (int-6137)
Using Heron's formula to find the area of a triangle (int-6475)

Exercise 6.2 Area

learnon

6.2 Quick quiz on | 6.2 Exercise

Individual pathways

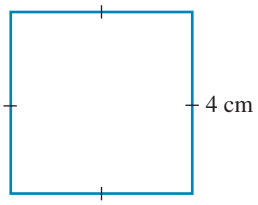
PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 9, 11, 14, 15, 19, 22	2, 5, 8, 12, 16, 17, 20, 23	3, 6, 10, 13, 18, 21, 24

Unless told otherwise, where appropriate, give answers correct to 2 decimal places.

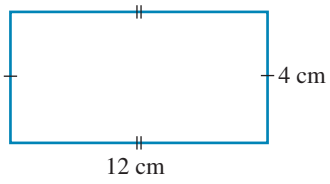
Fluency

1. Calculate the areas of the following shapes.

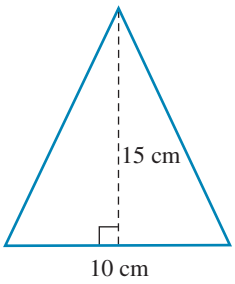
a.



b.

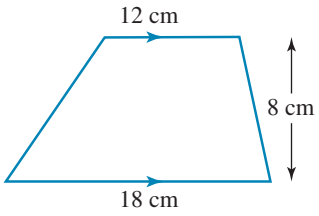


c.

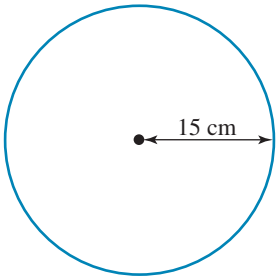


2. Calculate the areas of the following shapes.

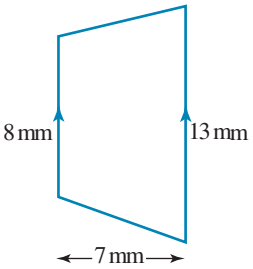
a.



b.

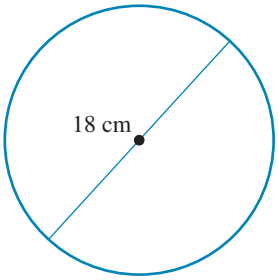


c.

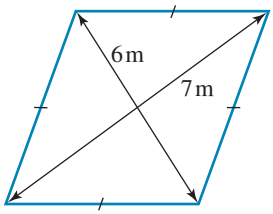


3. Calculate the areas of the following shapes.

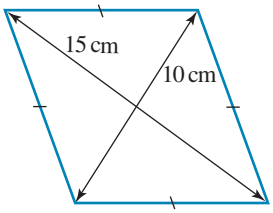
a.



b.

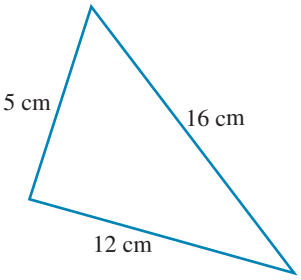


c.

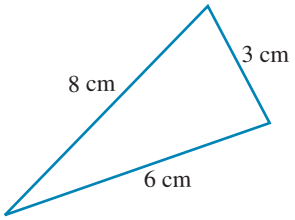


4. WE1a Use Heron's formula to calculate the areas of the following triangles, correct to 2 decimal places.

a.

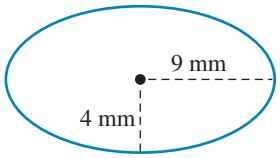


b.

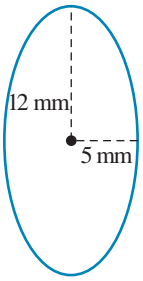


5. WE1b Calculate the areas of the following ellipses, correct to 1 decimal place.

a.



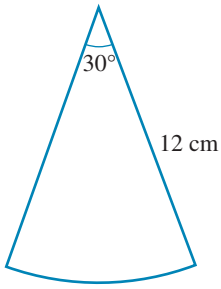
b.



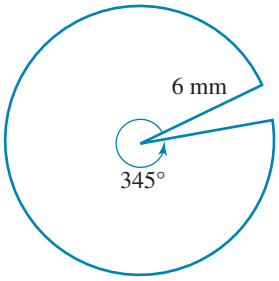
6. WE1c Calculate the area of each of the following shapes:

i. stating the answer exactly, that is in terms of π
ii. correct to 2 decimal places.

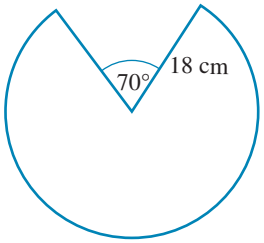
a.



b.



c.

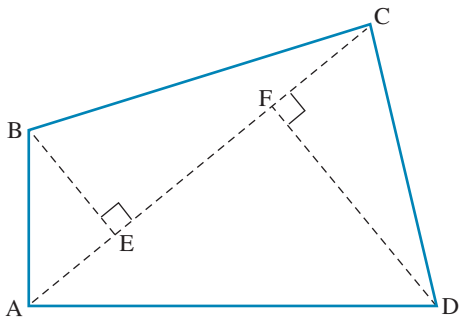


7. MC A figure has an area of about 64 cm^2. Identify which of the following *cannot* possibly represent the figure.

A. A triangle with base length 16 cm and height 8 cm
B. A circle with radius 4.51 cm
C. A rectangle with dimensions 16 cm and 4 cm
D. A rhombus with diagonals 16 cm and 4 cm

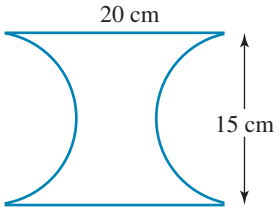
8. MC Identify from the following list all the lengths required to calculate the area of the quadrilateral shown.

A. AB, BC, CD and AD
B. AB, BE, AC and CD
C. BC, BE, AD and CD
D. AC, BE and FD

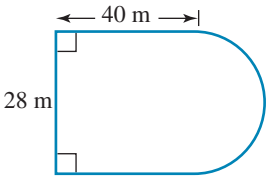


9. WE2 Calculate the areas of the following composite shapes.

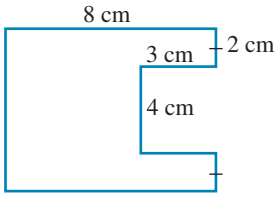
a.



b.

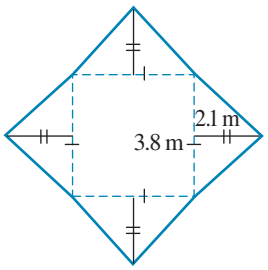


c.

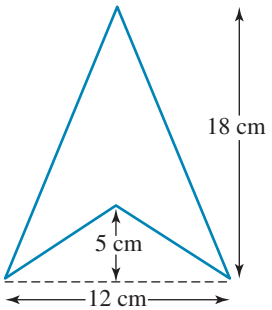


10. Calculate the areas of the following composite shapes.

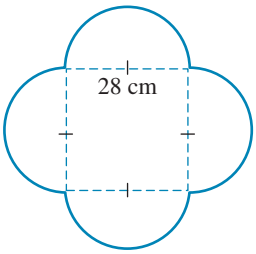
a.



b.

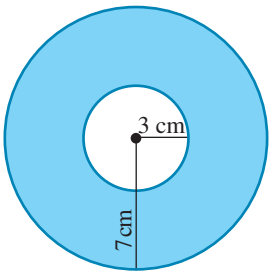


c.

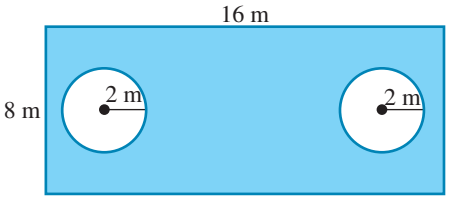


11. Calculate the shaded area in each of the following.

a.

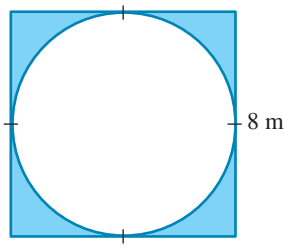


b.

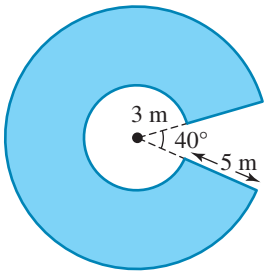


12. Calculate the shaded area in each of the following.

a.

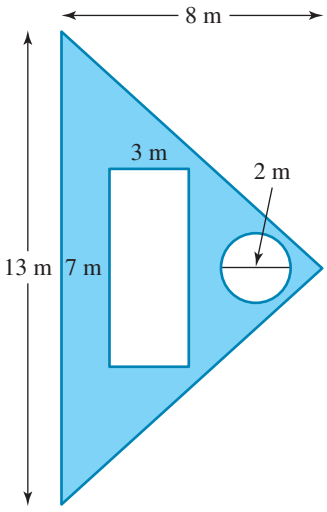


b.

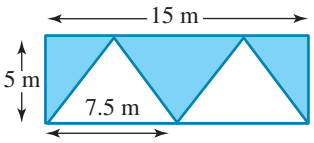


13. Calculate the shaded area in each of the following.

a.



b.



Understanding

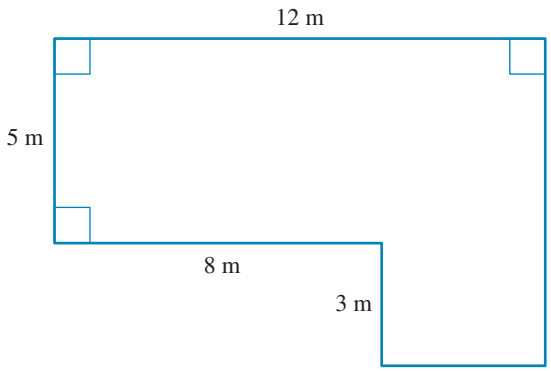
14. A sheet of cardboard is 1.6 m by 0.8 m. The following shapes are cut from the cardboard:
- a circular piece with radius 12 cm
- a rectangular piece 20 cm by 15 cm
- two triangular pieces with base length 30 cm and height 10 cm
- a triangular piece with side lengths 12 cm, 10 cm and 8 cm.

Calculate the area of the remaining piece of cardboard.

15. A rectangular block of land, 12 m by 8 m, is surrounded by a concrete path 0.5 m wide. Calculate the area of the path.

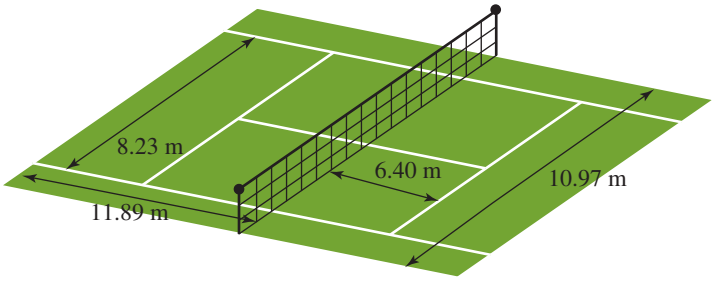
16. Concrete slabs 1 m by 0.5 m are used to cover a footpath 20 m by 1.5 m. Determine how many slabs are needed.

17. A city council builds a 0.5-m-wide concrete path around the garden as shown.



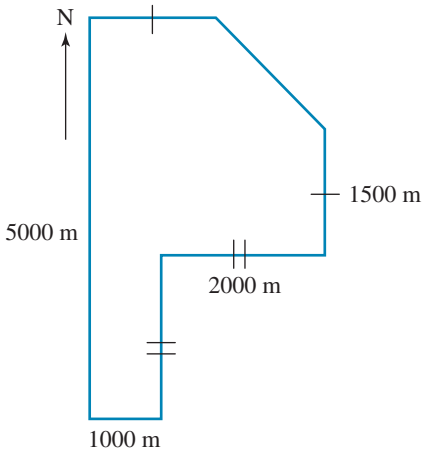
Determine the cost of the job if the worker charges \$40.00 per m^2.

18. A tennis court used for doubles is 10.97 m wide, but a singles court is only 8.23 m wide, as shown in the diagram.
- **a.** Calculate the area of the doubles court.
- **b.** Calculate the area of the singles court.
- **c.** Determine the percentage of the doubles court that is used for singles. Give your answer to the nearest whole number.



Reasoning

19. Dan has purchased a country property with layout and dimensions as shown in the diagram.
- **a.** Show that the property has a total area of 987.5 ha.
- **b.** Dan wants to split the property in half (in terms of area) by building a straight-line fence running either north–south or east–west through the property. Assuming the cost of the fencing is a fixed amount per linear metre, justify where the fence should be built (that is, how many metres from the top left-hand corner and in which direction) to minimise the cost.



20. Ron the excavator operator has 100 metres of barricade mesh and needs to enclose an area to safely work in. He chooses to make a rectangular region with dimensions x and y. Show your working when required.

- **a.** Write an equation that connects x, y and the perimeter.
- **b.** Write y in terms of x.
- **c.** Write an equation for the area of the region in terms of x.
- **d.** Fill in the table for different values of x.

x	0	5	10	15	20	25	30	35	40	45	50
Area (m^2)											

- **e.** Can x have a value more than 50? Why?
- **f.** Sketch a graph of area against x.
- **g.** Determine the value of x that makes the area a maximum.
- **h.** Determine the value of y for maximum area.
- **i.** Determine the shape that encloses the maximum area.
- **j.** Calculate the maximum area.

Ron decides to choose to make a circular area with the barricade mesh.

- **k.** Calculate the radius of this circular region.
- **l.** Calculate the area that is enclosed in this circular region.
- **m.** Determine how much extra area Ron now has compared to his rectangular region.

21. In question **20**, Ron the excavator operator could choose to enclose a rectangular or circular area with 150 m of barricade mesh. In this case, the circular region resulted in a larger safe work area.

- **a.** Show that for 150 m of barricade mesh, a circular region again results in a larger safe work area as opposed to a rectangular region.
- **b.** Show that for n metres of barricade mesh, a circular region will result in a larger safe work area as opposed to a rectangular region.

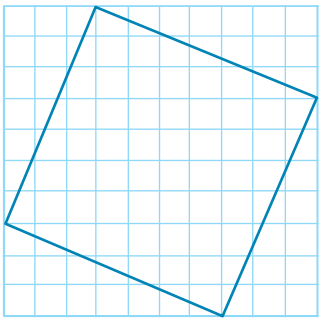
Problem solving

22. A vegetable gardener is going to build four new rectangular garden beds side by side. Each garden bed measures 12.5 metres long and 3.2 metres wide. To access the garden beds, the gardener requires a path 1 metre wide between each garden bed and around the outside of the beds.

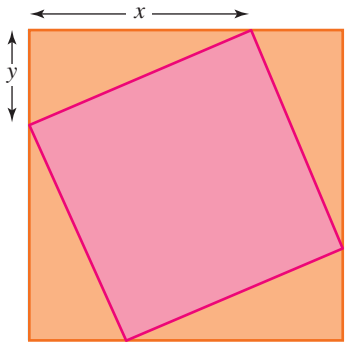
- **a.** Evaluate the total area the vegetable gardener would need for the garden beds and paths.
- **b.** The garden beds need to be mulched. A bag of mulch costing $29.50 covers an area of 25 square metres. Determine how many bags of mulch the gardener will need to purchase.
- **c.** The path is to be resurfaced at a cost of $39.50 per 50 square metres. Evaluate the cost of resurfacing the path.
- **d.** The gardener needs to spend a further $150 on plants. Determine the total cost of building these new garden beds and paths.

23. The diagram shows one smaller square drawn inside a larger square on grid paper.

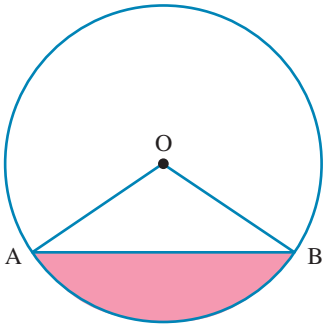
- **a.** Determine what fraction of the area of the larger square is the area of the smaller square.



b. Another square with side lengths of 10 cm has a smaller square drawn inside. Determine the values of x and y if the smaller square is half the larger square.



24. The shaded area in the diagram is called a segment of a circle.
A circle with a radius of 10 cm has ∠AOB equal to 90°.
A second circle, also with a radius of 10 cm, has ∠AOB equal to 120°.
Evaluate the difference in the areas of the segments of these two circles, correct to 2 decimal places.



LESSON
6.3 Total surface area

LEARNING INTENTIONS

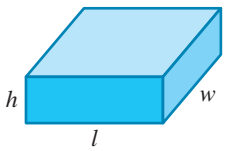
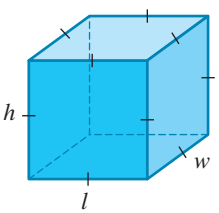
At the end of this lesson you should be able to:
- calculate the total surface areas of rectangular prisms and pyramids
- calculate the total surface areas of cylinders and spheres
- calculate the total surface areas of cones

6.3.1 Total surface areas of solids

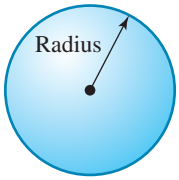
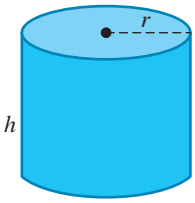
eles-4811

- The **total surface area (TSA)** of a solid is the sum of the areas of all the faces of that solid.

TSAs of rectangular prisms and cubes

Shape	Diagram	Formula
Rectangular prism (cuboid)		$TSA = 2(lh + lw + wh)$
Cube		$TSA = 6l^2$

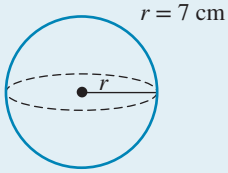
TSAs of spheres and cylinders

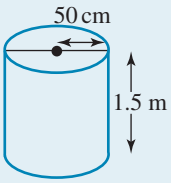
Shape	Diagram	Formula
Sphere		$\text{TSA} = 4\pi r^2$
Cylinder		$\text{TSA} = A_{\text{curved surface}} + A_{\text{circular ends}}$ $= 2\pi rh + 2\pi r^2$ $= 2\pi r(h + r)$

Note: Total surface area of pyramids, cones and spheres is part of the 10A course.

WORKED EXAMPLE 3 Calculating TSAs of solids

Calculate the total surface area of the solids, correct to the nearest cm^2.

a. 

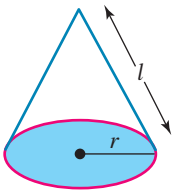
b. 

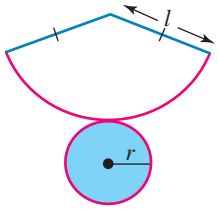
THINK	WRITE
a. 1. Write the formula for the TSA of a sphere.	**a.** $\text{TSA} = 4\pi r^2$
2. Identify the value for r.	$r = 7$
3. Substitute and evaluate.	$\text{TSA} = 4 \times \pi \times 7^2$ $\approx 615.8\ \text{cm}^2$
4. Write the answer to correct to the nearest cm^2.	$\approx 616\ \text{cm}^2$
b. 1. Write the formula for the TSA of a cylinder.	**b.** $\text{TSA} = 2\pi r(r + h)$
2. Identify the values for r and h. Note that the units will need to be the same.	$r = 50\ \text{cm}$, $h = 1.5\ \text{m}$ $= 150\ \text{cm}$
3. Substitute and evaluate.	$\text{TSA} = 2 \times \pi \times 50 \times (50 + 150)$ $= 62\,831.9\ \text{cm}^2$
4. Write the answer to correct to the nearest cm^2.	$\approx 62\,832\ \text{cm}^2$

6.3.2 Total surface areas of cones

eles-4812

- The total surface area of a cone can be found by considering its net, which is comprised of a small circle and a sector of a larger circle.

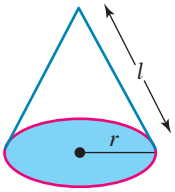




r = radius of the cone
l = slant height of the cone

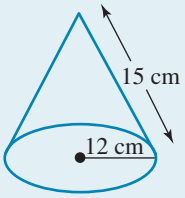
- The sector is a fraction of the full circle of radius l with circumference $2\pi l$.
- The sector has an arc length equivalent to the circumference of the base of the cone, $2\pi r$.
- The fraction of the full circle represented by the sector can be found by writing the arc length as a fraction of the circumference of the full circle, $\frac{2\pi r}{2\pi l} = \frac{r}{l}$.

$$\begin{aligned}\text{Area of a sector} &= \text{fraction of the circle} \times \pi l^2 \\ &= \frac{r}{l} \times \pi l^2 \\ &= \pi r l\end{aligned}$$

Shape	Diagram	Formula
Cone		$\text{TSA} = A_{\text{curved surface}} + A_{\text{circular end}}$ $= \pi r^2 + \pi r l$ $= \pi r(r + l)$

WORKED EXAMPLE 4 Calculating the TSA of a cone

Calculate the total surface area of the cone shown.

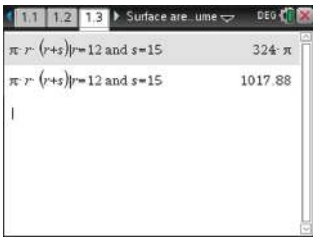


THINK	WRITE
1. Write the formula for the TSA of a cone.	$\text{TSA} = \pi r(r + l)$
2. State the values of r and l.	$r = 12, l = 15$
3. Substitute and evaluate to obtain the answer.	$\text{TSA} = \pi \times 12 \times (12 + 15)$ $= 1017.9\ \text{cm}^2$

TI | THINK

On the Calculator page, complete the entry line as:
$\pi\,(r+s)\,|\,r=12$
and $s=15$
Press CTRL ENTER to get a decimal approximation.

DISPLAY/WRITE

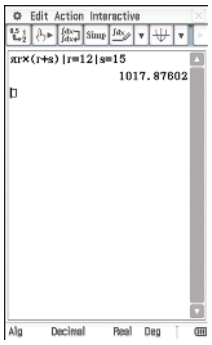


The total surface area of the cone is 1017.9 cm^2 correct to 1 decimal place.

CASIO | THINK

On the Main screen in decimal mode, complete the entry lines as:
$\pi r\,(r+s)\,|\,r=12\,|\,s=15$
Then press EXE.

DISPLAY/WRITE



The total surface area of the cone is 1017.9 cm^2 correct to 1 decimal place.

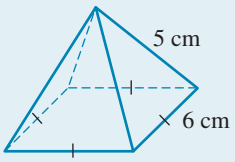
eles-4813

6.3.3 Total surface areas of other solids

- The TSA of a solid can be found by summing the areas of each face.
- Check the total number of faces to ensure that none are left out.

WORKED EXAMPLE 5 Calculating the TSA of a pyramid

Calculate the total surface area of the square-based pyramid shown.



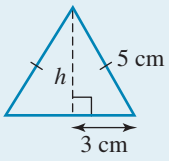
THINK

1. There are five faces: the square base and four identical triangles.
2. Calculate the area of the square base.
3. Draw and label one triangular face and write the formula for determining its area.
4. Calculate the height of the triangle, h, using Pythagoras' theorem.

WRITE/DRAW

TSA = area of square base + area of four triangular faces

Area of base = l^2, where $l = 6$
Area of base = 6^2
$= 36\ cm^2$



Area of a triangular face = $\frac{1}{2}bh$; $b = 6$

$a^2 = c^2 - b^2$, where $a = h$, $b = 3$, $c = 5$
$h^2 = 5^2 - 3^2$
$h^2 = 25 - 9$
$h^2 = 16$
$h = 4$ cm

5. Calculate the area of the triangular face by substituting $b = 6$ and $h = 4$.

$$\text{Area of triangular face} = \frac{1}{2} \times 6 \times 4$$
$$= 12\,\text{cm}^2$$

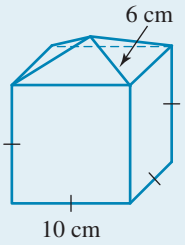
6. Calculate the TSA by adding the area of the square base and the area of the four identical triangular faces together.

$$\text{TSA} = 36 + 4 \times 12$$
$$= 36 + 48$$
$$= 84\,\text{cm}^2$$

Note: The area of the triangular faces can be found using Heron's formula. This method is demonstrated in the following worked example.

WORKED EXAMPLE 6 Calculating the TSA of a solid

Calculate the total surface area of the solid shown correct to 1 decimal place.

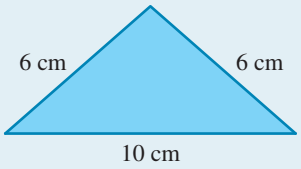


THINK

1. The solid shown has nine faces — five identical squares and four identical triangles.
2. Calculate the area of one square face with the side length 10 cm.
3. Draw a triangular face and label the three sides. Use Heron's formula to calculate the area.
4. State the formula for s, the semi-perimeter. Substitute the values of a, b and c and evaluate the value of s.
5. State Heron's formula for the area of one triangle. Substitute and evaluate.

WRITE/DRAW

$$\text{TSA} = 5 \times \text{area of a square} + 4 \times \text{area of a triangle}$$
$$A_{\text{square}} = l^2, \text{ where } l = 10$$
$$A = 10^2$$
$$A = 100\,\text{cm}^2$$



$$s = \frac{a + b + c}{2}$$
$$s = \frac{6 + 6 + 10}{2}$$
$$s = 11$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$
$$A = \sqrt{11(11-6)(11-6)(11-10)}$$
$$A = \sqrt{275}$$
$$A = 16.583\,124\ldots\,\text{cm}^2$$

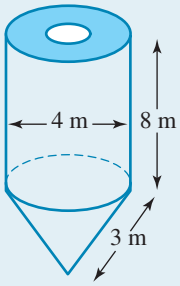
6. Determine the TSA of the solid by adding the area of the five squares and four triangles.

$$\begin{aligned}\text{TSA} &= 5 \times 100 + 4 \times 16.583\,124\ldots \\ &= 566.3325\ldots \\ &= 566.3\ \text{cm}^2 \text{ (to 1 decimal place)}\end{aligned}$$

Note: Rounding is not done until the final step. It is important to realise that rounding too early can affect the accuracy of results.

WORKED EXAMPLE 7 Applying surface area in worded problems

The silo shown is to be built from metal. The top portion of the silo is a cylinder of diameter 4 m and height 8 m. The bottom part of the silo is a cone of slant height 3 m. The silo has a circular opening of radius 30 cm on the top.





a. Calculate the area of metal (to the nearest m^2) that is required to build the silo.
b. If it costs $12.50 per m^2 to cover the surface with an anti-rust material, determine how much will it cost to cover the silo completely.

THINK

a. 1. The surface area of the silo consists of an annulus, the curved part of the cylinder and the curved section of the cone.

2. To calculate the area of the annulus, subtract the area of the small circle from the area of the larger circle.
Let R = radius of small circle. Remember to convert all measurements to the same units.

3. The middle part of the silo is the curved part of a cylinder. Determine its area. (Note that in the formula $\text{TSA}_{\text{cylinder}} = 2\pi r^2 + 2\pi rh$, the curved part is represented by $2\pi rh$.)

4. The bottom part of the silo is the curved section of a cone. Determine its area. (Note that in the formula $\text{TSA}_{\text{cone}} = \pi r^2 + \pi rl$, the curved part is given by πrl.)

WRITE

a. TSA = area of annulus
+ area of curved section of a cylinder
+ area of curved section of a cone

$$\begin{aligned}\text{Area of annulus} &= A_{\text{large circle}} - A_{\text{small circle}} \\ &= \pi r^2 - \pi R^2\end{aligned}$$

where $r = \frac{4}{2} = 2$ m and $R = 30$ cm $= 0.3$ m.

$$\begin{aligned}\text{Area of annulus} &= \pi \times 2^2 - \pi \times 0.3^2 \\ &= 12.28\ \text{m}\end{aligned}$$

$$\text{Area of curved section of cylinder} = 2\pi rh$$

where $r = 2$, $h = 8$.

$$\begin{aligned}\text{Area of curved section of cylinder} &= 2 \times \pi \times 2 \times 8 \\ &= 100.53\ \text{m}^2\end{aligned}$$

$$\text{Area of curved section of cone} = \pi rl$$

where $r = 2$, $l = 3$.

$$\begin{aligned}\text{Area of curved section of cone} &= \pi \times 2 \times 3 \\ &= 18.85\ \text{m}^2\end{aligned}$$

5. Calculate the total surface area of the silo by finding the sum of the surface areas calculated above.
6. Write the answer in words.

$\text{TSA} = 12.28 + 100.53 + 18.85$
$= 131.66\ \text{m}^2$

The area of metal required is $132\ \text{m}^2$, correct to the nearest square metre.

b. To determine the total cost, multiply the total surface area of the silo by the cost of the anti-rust material per m^2 ($12.50).

b. $\text{Cost} = 132 \times \$12.50$
$= \$1650.00$

Resources

eWorkbook	Topic 6 workbook (worksheets, code puzzle and a project) (ewbk-13290)
Video eLesson	Total surface areas of prisms (eles-1909)
Interactivities	Individual pathway interactivity: Total surface area (int-4594) Surface area of a prism (int-6079) Surface area of a cylinder (int-6080) Surface area (int-6477)

Exercise 6.3 Total surface area

learnon

6.3 Quick quiz on | **6.3 Exercise**

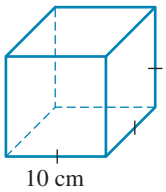
Individual pathways

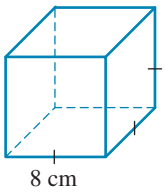
PRACTISE	CONSOLIDATE	MASTER
1, 5, 7, 10, 12, 17, 20	2, 6, 9, 11, 14, 15, 18, 21	3, 4, 8, 13, 16, 19, 22

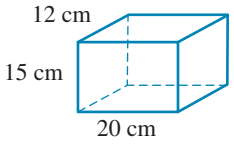
Unless told otherwise, where appropriate, give answers correct to 1 decimal place.

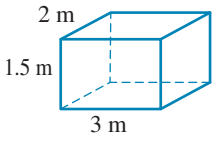
Fluency

1. Calculate the total surface areas of the solids shown.

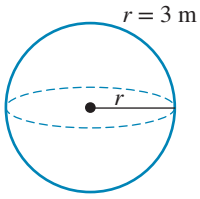
a.


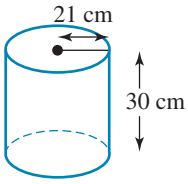
b.


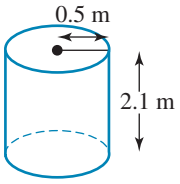
c.


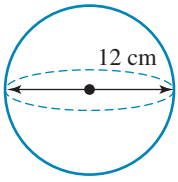
d.


2. WE3 Calculate the total surface areas of the solids shown below.

a.


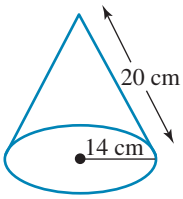
b.


c.


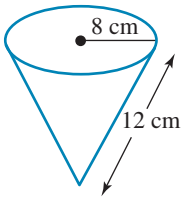
d.


3. WE4 Calculate the total surface areas of the cones below.

a.

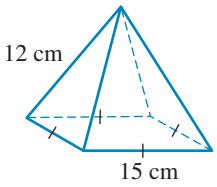


b.

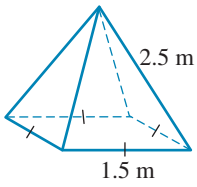


4. WE5 Calculate the total surface areas of the solids below.

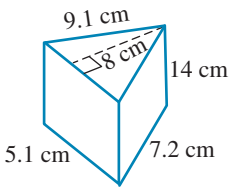
a.



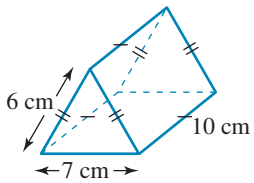
b.



c.



d.



5. Calculate the surface areas of the following.

a. A cube of side length 1.5 m

b. A rectangular prism 6 m × 4 m × 2.1 m

c. A cylinder of radius 30 cm and height 45 cm, open at one end

6. Calculate the surface areas of the following.

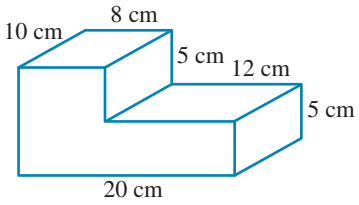
a. A sphere of radius 28 mm

b. An open cone of radius 4 cm and slant height 10 cm

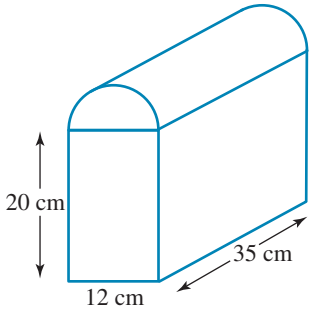
c. A square pyramid of base length 20 cm and slant edge 30 cm

7. WE6 Calculate the total surface areas of the objects shown.

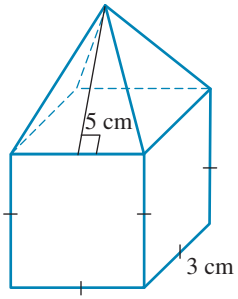
a.



b.

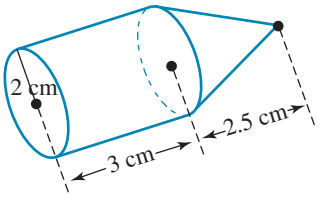


c.

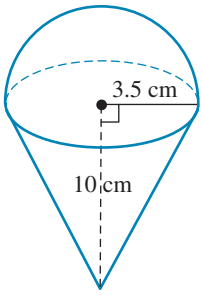


8. Calculate the total surface areas of the objects shown.

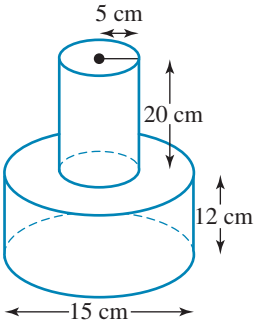
a.



b.



c.



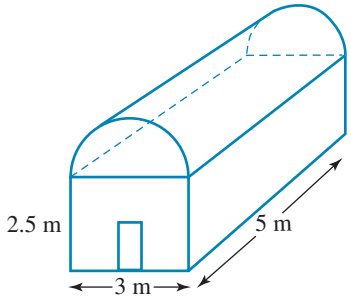
9. MC A cube has a total surface area of 384 cm^2. Calculate the length of the edge of the cube.

A. 9 cm **B.** 8 cm **C.** 7 cm **D.** 6 cm

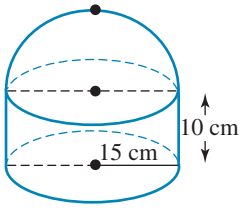
Understanding

10. **WE7** The greenhouse shown is to be built using shade cloth. It has a wooden door of dimensions $1.2\,\text{m} \times 0.5\,\text{m}$.

a. Calculate the total area of shade cloth needed to complete the greenhouse.
b. Determine the cost of the shade cloth at \$6.50 per m^2.



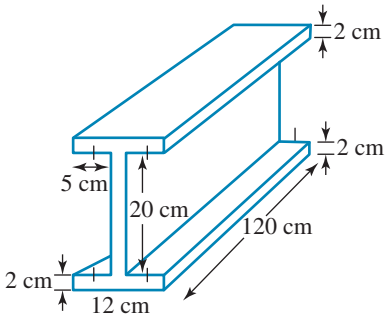
11. A cylinder is joined to a hemisphere to make a cake holder, as shown.



The surface of the cake holder is to be chromed at 5.5 cents per cm^2.

a. Calculate the total surface area to be chromed.
b. Determine the cost of chroming the cake holder.

12. A steel girder is to be painted. Calculate the area of the surface to be painted.

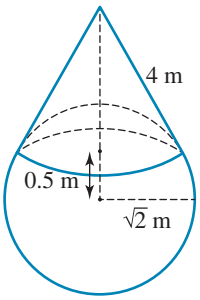


13. Open cones are made from nets cut from a large sheet of paper $1.2\,\text{m} \times 1.0\,\text{m}$. If a cone has a radius of 6 cm and a slant height of 10 cm, determine how many cones can be made from the sheet.
(Assume there is 5% wastage of paper.)

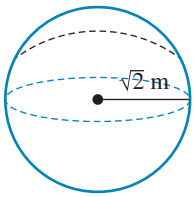
14. A prism of height 25 cm has a base in the shape of a rhombus with diagonals of 12 cm and 16 cm. Calculate the total surface area of the prism.

15. A hemispherical glass dome, with a diameter of 24 cm, sits on a concrete cube with sides of 50 cm. To protect the structure, all exposed sides are to be treated. The glass costs \$1.50/cm^2 to treat and the concrete costs 5c/cm^2.
Calculate the cost in treating the structure if the base of the cube is already fixed to the ground. Give your answer to the nearest dollar.

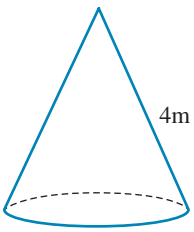
16. An inverted cone with side length 4 metres is placed on top of a sphere such that the centre of the cone's base is 0.5 metres above the centre of the sphere. The radius of the sphere is $\sqrt{2}$ metres.



a. Calculate the exact total surface area of the sphere.



b. Calculate the radius of the cone exactly.



c. Calculate the area of the curved surface of the cone exactly.

Reasoning

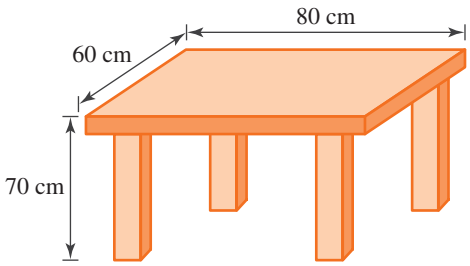
17. A shower recess with dimensions 1500 mm (back wall) by 900 mm (side wall) needs to have the back and two side walls tiled to a height of 2 m.

a. Calculate the area to be tiled in m^2.

b. Justify that 180 tiles (including those that need to be cut) of dimension 20 cm by 20 cm will be required. Disregard the grout and assume that once a tile is cut, only one piece of the tile can be used.

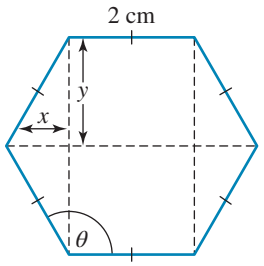
c. Evaluate the cheapest option of tiling; \$1.50/tile or \$39.50/box, where a box covers 1 m^2, or tiles of dimension 30 cm by 30 cm costing \$3.50/tile.

18. The table shown below is to be varnished (including the base of each leg). The tabletop has a thickness of 180 mm and the cross-sectional dimensions of the legs are 50 mm by 50 mm. A friend completes the calculation without a calculator as shown. Assume there are no simple calculating errors. Analyse the working presented and justify if the TSA calculated is correct.



Tabletop (inc. leg bases)	0.96	$2 \times (0.8 \times 0.6)$
Legs	0.416	$16 \times (0.52 \times 0.05)$
Tabletop edging	0.504	$0.18 \times (2\,(0.8 + 0.6))$
TSA	1.88 m^2	

19. A soccer ball is made up of a number of hexagons sewn together on its surface. Each hexagon can be considered to have dimensions as shown in the diagram.



a. Calculate $\theta°$.
b. Calculate the values of x and y exactly.
c. Calculate the area of the trapezium in the diagram.
d. Hence, determine the area of the hexagon.
e. If the total surface area of the soccer ball is $192\sqrt{3}\,\text{cm}^2$, determine how many hexagons are on its surface.

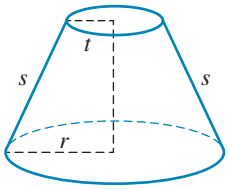
Problem solving

20. Tina is re-covering a footstool in the shape of a cylinder with diameter 50 cm and height 30 cm. She also intends to cover the base of the cushion. She has 1 m^2 of fabric to make this footstool.
When calculating the area of fabric required, allow an extra 20% of the total surface area to cater for seams and pattern placings.
Explain whether Tina has enough material to cover the footstool.

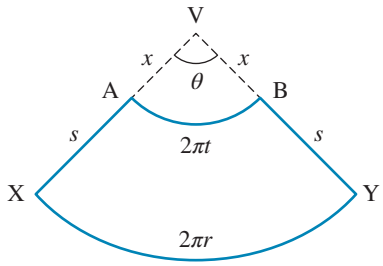


21. If the surface area of a sphere to that of a cylinder is in the ratio 4 : 3 and the sphere has a radius of $3a$, show that if the radius of the cylinder is equal to its height, then the radius of the cylinder is $\dfrac{3\sqrt{3}a}{2}$.

22. A frustum of a cone is a cone with the top sliced off, as shown.



When the curved side is 'opened up', it creates a shape, ABYX, as shown in the diagram.



a. Write an expression for the arc length XY in terms of the angle θ. Write another expression for the arc length AB in terms of the same angle θ. Show that, in radians, $\theta = \dfrac{2\pi(r-t)}{s}$.

b. **i.** Using the above formula for θ, show that $x = \dfrac{st}{(r-t)}$.

ii. Use similar triangles to confirm this formula.

c. Determine the areas of sectors AVB and XVY and hence determine the area of ABYX. Add the areas of the 2 circles to the area of ABYX to determine the TSA of a frustum.

LESSON
6.4 Volume

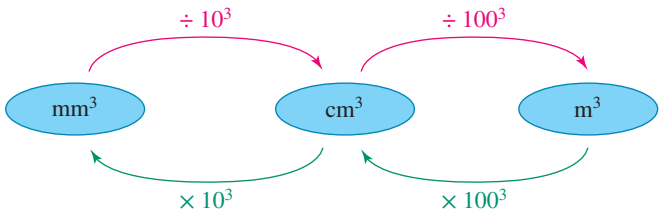
LEARNING INTENTIONS

At the end of this lesson you should be able to:
- calculate the volumes of prisms, including cylinders
- calculate the volumes of spheres
- calculate the volumes of pyramids.

6.4.1 Volume

eles-4814

- The **volume** of a 3-dimensional object is the amount of space it takes up.
- Volume is measured in units of mm^3, cm^3 and m^3.
- The following diagram can be used to convert between units of volume.



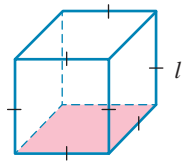
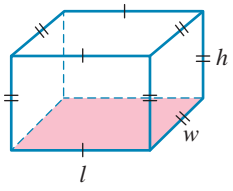
Volume of a prism

- The volume of any solid with a uniform cross-sectional area is given by the formula shown below.

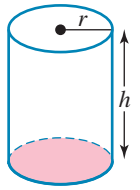
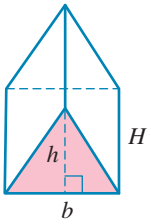
Volume of a solid with uniform cross-sectional area

$$V = AH$$

where A is the area of the cross-section and H is the height of the solid.

Shape	Diagram	Formula
Cube		Volume $= AH$ $=$ area of a square $\times$ height $= l^2 \times l$ $= l^3$
Rectangular prism		Volume $= AH$ $=$ area of a rectangle $\times$ height $= lwh$

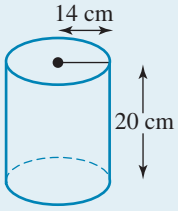
(continued)

Shape	Diagram	Formula
Cylinder		Volume $= AH$ $=$ area of a circle $\times$ height $= \pi r^2 h$
Triangular prism		Volume $= AH$ $=$ area of a triangle $\times$ height $= \frac{1}{2}bh \times H$

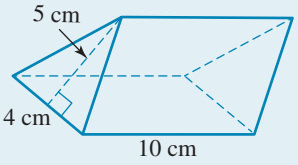
WORKED EXAMPLE 8 Calculating volumes of prisms

Calculate the volumes of the following shapes.

a.



b.



THINK

a. 1. Write the formula for the volume of the cylinder (prism).

2. Identify the value of the pronumerals.

3. Substitute and evaluate the answer.

b. 1. Write the formula for the volume of a triangular prism.

2. Identify the value of the pronumerals. *Note:* h is the height of the triangle and H is the depth of the prism.

3. Substitute and evaluate the answer.

WRITE

a. $V = AH$

$= \pi r^2 h$

$r = 14,\ h = 20$

$V = \pi \times 14^2 \times 20$

$\approx 12\,315.04\ \text{cm}^3$

b. $V = \frac{1}{2}bh \times H$

$b = 4,\ h = 5,\ H = 10$

$V = \frac{1}{2} \times 4 \times 5 \times 10$

$= 100\ \text{cm}^3$

WORKED EXAMPLE 9 Changing the dimensions of a prism

a. If each of the side lengths of a cube are doubled, then determine the effect on its volume.

b. If the radius is halved and the height of a cylinder is doubled, then determine the effect on its volume.

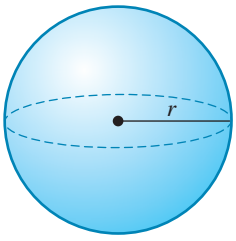
THINK	WRITE
a. 1. Write the formula for the volume of the cube.	**a.** $V = l^3$
2. Identify the value of the pronumeral. *Note:* Doubling is the same as multiplying by 2.	$l_{\text{new}} = 2l$
3. Substitute and evaluate.	$V_{\text{new}} = (2l)^3$
4. Compare the answer obtained in step 3 with the volume of the original shape.	$= 8l^3$
5. Write your answer.	Doubling each side length of a cube increases the volume by a factor of 8; that is, the new volume will be 8 times as large as the original volume.
b. 1. Write the formula for the volume of the cylinder.	**b.** $V = \pi r^2 h$
2. Identify the value of the pronumerals. *Note:* Halving is the same as dividing by 2.	$r_{\text{new}} = \frac{r}{2}$, $h_{\text{new}} = 2h$
3. Substitute and evaluate.	$V_{\text{new}} = \pi \left(\frac{r}{2}\right)^2 2h$
	$= \pi \times \frac{r^2}{2^{\cancel{4}}} \times \cancel{2}h$
	$= \frac{\pi r^2 h}{2}$
4. Compare the answer obtained in step 3 with the volume of the original shape.	$= \frac{1}{2}\pi r^2 h$
5. Write your answer.	Halving the radius and doubling the height of a cylinder decreases the volume by a factor of 2; that is, the new volume will be half the original volume.

eles-4815

6.4.2 Volumes of common shapes

Volume of a sphere

- The volume of a sphere of radius r is given by the following formula.

Shape	Diagram	Formula
Sphere		$V = \frac{4}{3}\pi r^3$

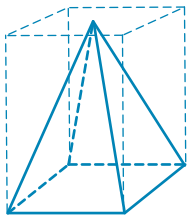
WORKED EXAMPLE 10 Calculating the volume of a sphere

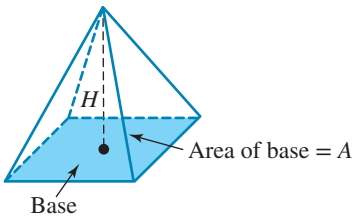
Find the volume of a sphere of radius 9 cm. Answer correct to 1 decimal place.

THINK	WRITE
1. Write the formula for the volume of a sphere.	$V = \frac{4}{3}\pi r^3$
2. Identify the value of r.	$r = 9$
3. Substitute and evaluate.	$V = \frac{4}{3} \times \pi \times 9^3$ $= 3053.6\,\text{cm}^3$

Volume of a pyramid

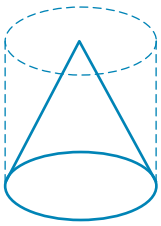
- Pyramids are not prisms, as the cross-section changes from the base upwards.
- The volume of a pyramid is one-third the volume of the prism with the same base and height.

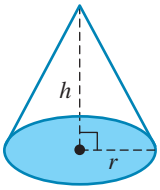


Shape	Diagram	Formula
Pyramid		$V_{\text{pyramid}} = \frac{1}{3}AH$

Volume of a cone

- The cone is a pyramid with a circular base.



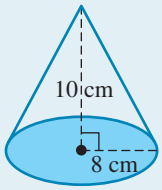
Shape	Diagram	Formula
Cone		$V_{cone} = \frac{1}{3}\pi r^2 h$

Note: Volume of cones, pyramids and spheres is part of the 10A course.

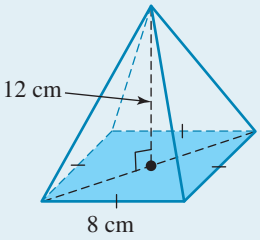
WORKED EXAMPLE 11 Calculating the volume of pyramids and cones

Calculate the volume of each of the following solids.

a.



b.



THINK	WRITE
a. 1. Write the formula for the volume of a cone.	**a.** $V = \frac{1}{3}\pi r^2 h$
2. Identify the values of r and h.	$r = 8,\ h = 10$
3. Substitute and evaluate.	$V = \frac{1}{3} \times \pi \times 8^2 \times 10$ $= 670.21 \text{ cm}^3$
b. 1. Write the formula for the volume of a pyramid.	**b.** $V = \frac{1}{3}AH$
2. Calculate the area of the square base.	$A = l^2$ where $l = 8$ $A = 8^2$ $= 64 \text{ cm}^2$
3. Identify the value of H.	$H = 12$
4. Substitute and evaluate.	$V = \frac{1}{3} \times 64 \times 12$ $= 256 \text{ cm}^3$

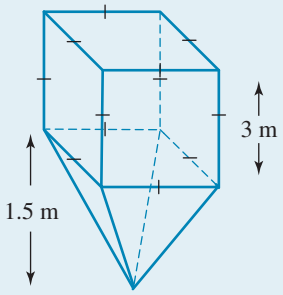
6.4.3 Volumes of composite solids

eles-4816

- A composite solid is a combination of a number of solids.
- Calculate the volume of each solid separately.
- Sum these volumes to give the volume of the composite solid.

WORKED EXAMPLE 12 Calculating the volume of a composite solid

Calculate the volume of the composite solid shown.



THINK	WRITE
1. The given solid is a composite figure, made up of a cube and a square-based pyramid.	$V = \text{Volume of cube} + \text{Volume of pyramid}$
2. Calculate the volume of the cube.	$V_{\text{cube}} = l^3$ where $l = 3$ $V_{\text{cube}} = 3^3$ $= 27\ \text{m}^3$
3. Write the formula for the volume of a square-based pyramid.	$V_{\text{square-based pyramid}} = \frac{1}{3}AH$
4. Calculate the area of the square base.	$A = l^2$ $= 3^2$ $= 9\ \text{m}^2$
5. Identify the value of H.	$H = 1.5$
6. Substitute and evaluate the volume of the pyramid.	$V_{\text{square-based pyramid}} = \frac{1}{3} \times 9 \times 1.5$ $= 4.5\ \text{m}^3$
7. Calculate the total volume by adding the volume of the cube and pyramid.	$V = 27 + 4.5$ $= 31.5\ \text{m}^3$

6.4.4 Capacity

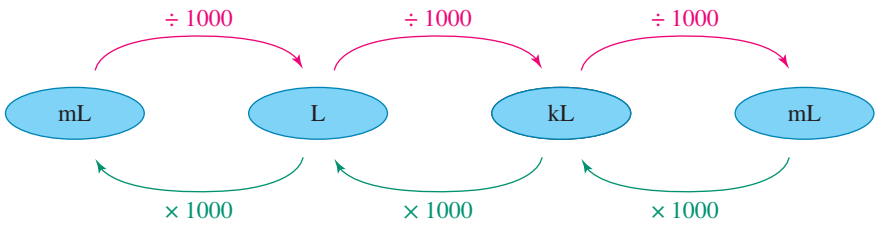
eles-4817

- Some 3-dimensional objects are hollow and can be filled with liquid or some other substance.
- The amount of substance that a container can hold is called its capacity.
- **Capacity** is essentially the same as volume but is usually measured in mL, L, kL and ML (megalitres) where $1\ \text{mL} = 1\ \text{cm}^3$

$1\ \text{L} = 1000\ \text{cm}^3$

$1\ \text{kL} = 1\ \text{m}^3$.

- The following diagram can be used to convert between units of capacity.



WORKED EXAMPLE 13 Calculating the capacity of a prism

Determine the capacity (in litres) of a cuboidal aquarium that is 50 cm long, 30 cm wide and 40 cm high.



THINK	WRITE
1. Write the formula for the volume of a rectangular prism.	$V = lwh$
2. Identify the values of the pronumerals.	$l = 50,\ w = 30,\ h = 40$
3. Substitute and evaluate.	$V = 50 \times 30 \times 40$ $= 60\,000\ \text{cm}^3$
4. State the capacity of the container in millilitres, using $1\ \text{cm}^3 = 1\ \text{mL}$.	$= 60\,000\ \text{mL}$
5. Since $1\ \text{L} = 1000\ \text{mL}$, to convert millilitres to litres divide by 1000.	$= 60\ \text{L}$
6. Write the answer in a sentence.	The capacity of the fish tank is 60 L.

on Resources

eWorkbook Topic 6 workbook (worksheets, code puzzle and a project) (ewbk-13290)

Interactivities Individual pathway interactivity: Volume (int-4595)
Volume 1 (int-3791)
Volume 2 (int-6476)
Volume of solids (int-3794)

Exercise 6.4 Volume

learnon

6.4 Quick quiz on | 6.4 Exercise

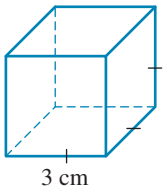
Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 5, 7, 9, 12, 15, 18, 23, 27	2, 6, 8, 10, 13, 16, 19, 21, 24, 28	4, 11, 14, 17, 20, 22, 25, 26, 29, 30

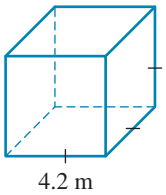
Fluency

1. Calculate the volumes of the following prisms.

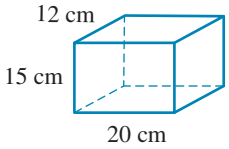
a.



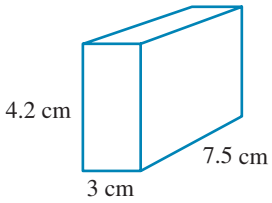
b.



c.

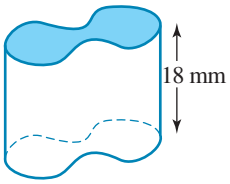


d.



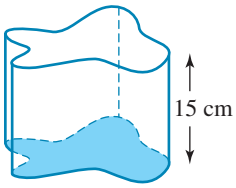
2. Calculate the volume of each of these solids.

a.



[Base area: 25 mm^2]

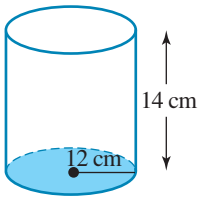
b.



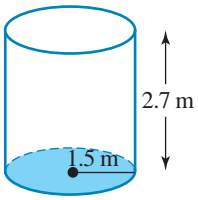
[Base area: 24 cm^2]

3. WE8 Calculate the volume of each of the following. Give each answer correct to 1 decimal place where appropriate.

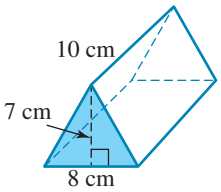
a.



b.

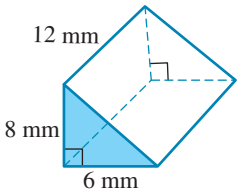


c.

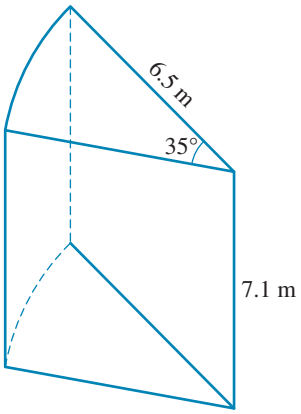


4. Calculate the volume of each of the following. Give each answer correct to 1 decimal place where appropriate.

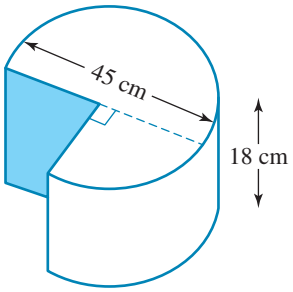
a.



b.



c.

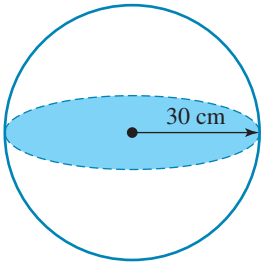


5. **WE10** Determine the volume of a sphere (correct to 1 decimal place) with a radius of:

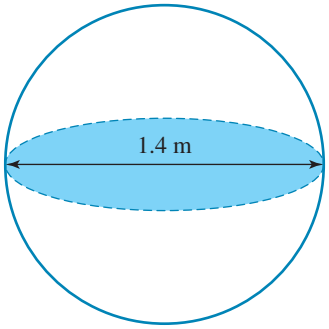
a. 1.2 m **b.** 15 cm **c.** 7 mm **d.** 50 cm

6. Calculate the volume of each of these figures, correct to 2 decimal places.

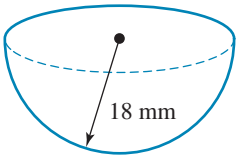
a.



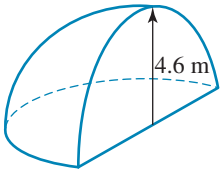
b.



c.

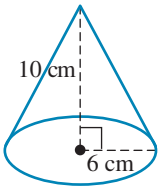


d.

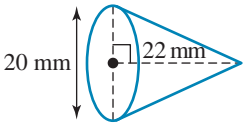


7. **WE11a** Determine the volume of each of the following cones, correct to 1 decimal place.

a.

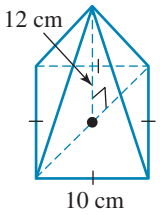


b.

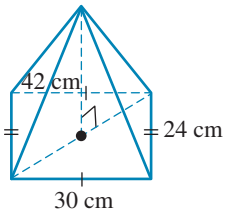


8. **WE11b** Calculate the volume of each of the following pyramids.

a.

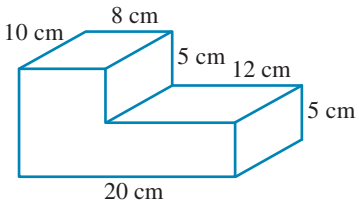


b.

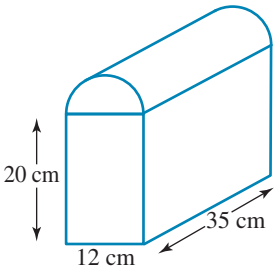


9. **WE12** Calculate the volume of each of the following composite solids correct to 2 decimal places where appropriate.

a.

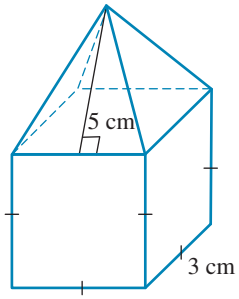


b.

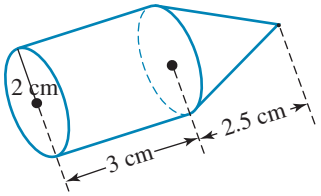


10. Calculate the volume of each of the following composite solids correct to 2 decimal places where appropriate.

a.

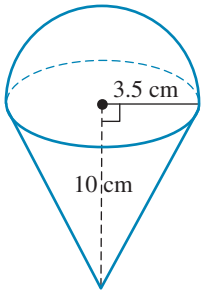


b.

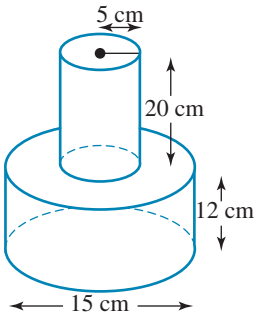


11. Calculate the volume of each of the following composite solids correct to 2 decimal places where appropriate.

a.



b.



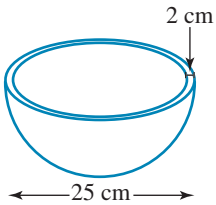
Understanding

12. WE9 Answer the following questions.

a. If the side length of a cube is tripled, then determine the effect on its volume.
b. If the side length of a cube is halved, then determine the effect on its volume.
c. If the radius is doubled and the height of a cylinder is halved, then determine the effect on its volume.
d. If the radius is doubled and the height of a cylinder is divided by four, then determine the effect on its volume.
e. If the length is doubled, the width is halved and the height of a rectangular prism is tripled, then determine the effect on its volume.



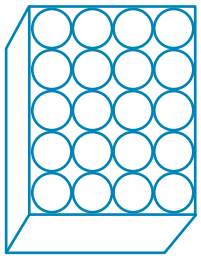
13. MC A hemispherical bowl has a thickness of 2 cm and an outer diameter of 25 cm.



If the bowl is filled with water, the capacity of the water will be closest to:

A. 1.526 L **B.** 1.30833 L **C.** 3.05208 L **D.** 2.42452 L

14. Tennis balls of diameter 8 cm are packed in a box 40 cm × 32 cm × 10 cm, as shown.



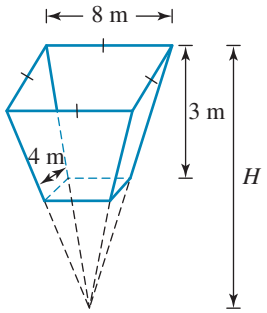
- **a.** Estimate the volume of one tennis ball.
  (*Hint:* Use π as approximately equal to 3.)
- **b.** Estimate the volume of all the tennis balls in the box.
- **c.** Estimate the volume of the box.
- **d.** Estimate the unfilled volume of the box.
- **e.** Determine, correct to 2 decimal places, how much space is left unfilled.
- **f.** Compare your estimate of the unfilled volume to your answer in part **e**.

15. **WE13** A cylindrical water tank has a diameter of 1.5 m and a height of 2.5 m. Determine the capacity (in litres) of the tank, correct to 1 decimal place.



16. A monument in the shape of a rectangular pyramid (base length of 10 cm, base width of 6 cm, height of 8 cm), a spherical glass ball (diameter of 17 cm) and conical glassware (radius of 14 cm, height of 10 cm) are packed in a rectangular prism of dimensions 30 cm by 25 cm by 20 cm. The extra space in the box is filled up by a packing material.
Determine, correct to 2 decimal places, the volume of packing material that is required.

17. A swimming pool is being constructed so that it is the upper part of an inverted square-based pyramid.



- **a.** Calculate H.
- **b.** Calculate the volume of the pool.
- **c.** Determine how many 6-m^3 bins will be required to take the dirt away.
- **d.** Determine how many litres of water are required to fill this pool.
- **e.** Determine how deep the pool is when it is half-filled.

18. A soft drink manufacturer is looking to repackage cans of soft drink to minimise the cost of packaging while keeping the volume constant. Consider a can of soft drink with a capacity of 400 mL.



- **a.** If the soft drink was packaged in a spherical can:
  - **i.** calculate the radius of the sphere, correct to 2 decimal places
  - **ii.** determine the total surface area of this can, correct to 1 decimal place.
- **b.** If the soft drink was packaged in a cylindrical can with a radius of 3 cm:
  - **i.** calculate the height of the cylinder, correct to 2 decimal places
  - **ii.** determine the total surface area of this can, correct to 2 decimal places.
- **c.** If the soft drink was packaged in a square-based pyramid with a base side length of 6 cm:
  - **i.** calculate the height of the pyramid, correct to 2 decimal places
  - **ii.** determine the total surface area of this can, correct to 2 decimal places.
- **d.** Explain which can you would recommend the soft drink manufacturer use for its repackaging.

19. The volume of a cylinder is given by the formula $V = \pi r^2 h$.

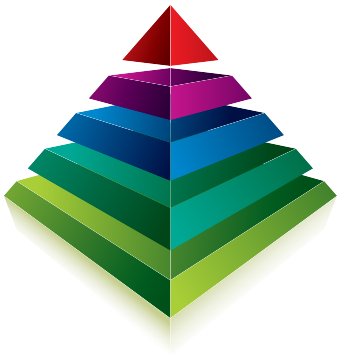
- **a.** Transpose the formula to make h the subject.
- **b.** A given cylinder has a volume of $1600\,\text{cm}^3$. Calculate its height, correct to 1 decimal place, if it has a radius of:
  - **i.** 4 cm
  - **ii.** 8 cm.
- **c.** Transpose the formula to make r the subject.
- **d.** Explain what restrictions must be placed on r.
- **e.** A given cylinder has a volume of $1800\,\text{cm}^3$. Determine its radius, correct to 1 decimal place, if it has a height of:
  - **i.** 10 cm
  - **ii.** 15 cm.

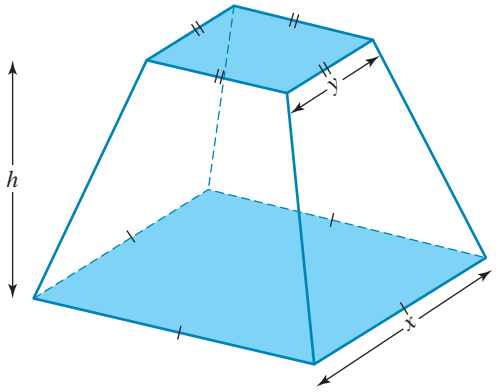
20. A toy maker has enough rubber to make one super-ball of radius 30 cm. Determine how many balls of radius 3 cm he can make from this rubber.

21. A manufacturer plans to make a cylindrical water tank to hold 2000 L of water.

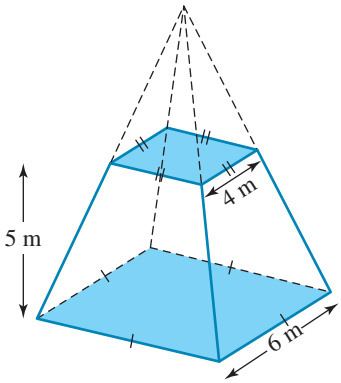
- **a.** Calculate the height, correct to 2 decimal places, if he uses a radius of 500 cm.
- **b.** Calculate the radius, correct to 2 decimal places if he uses a height of 500 cm.
- **c.** Determine the surface area of each of the two tanks. Assume the tank is a closed cylinder and give your answer in square metres correct to 2 decimal places.

22. The ancient Egyptians knew that the volume of the frustum of a square-based pyramid was given by the formula $V = \frac{1}{3}h\left(x^2 + xy + y^2\right)$, although how they discovered this is unclear. (A frustum is the part of a cone or pyramid that is left when the top is cut off.)





- **a.** Calculate the volume of the frustum below, correct to 2 decimal places.
- **b.** Determine the volume of the missing portion of the square-based pyramid shown, correct to 2 decimal places.

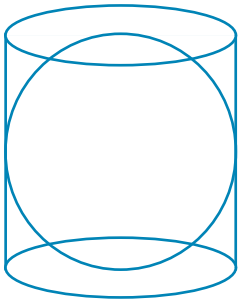


Reasoning

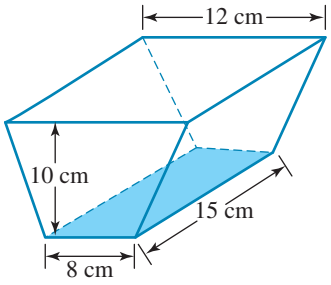
23. The Hastings' family house has a rectangular roof with dimensions 17 m × 10 m providing water to three cylindrical water tanks, each with a radius of 1.25 m and a height of 2.1 m.
Show that approximately 182 millimetres of rain must fall on the roof to fill the tanks.

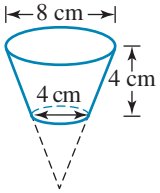
24. Archimedes is considered to be one of the greatest mathematicians of all time. He discovered several of the formulas used in this topic. Inscribed on his tombstone was a diagram of his proudest discovery.
It shows a sphere inscribed (fitting exactly) into a cylinder.
Show that:

$$\frac{\text{volume of the cylinder}}{\text{volume of the sphere}} = \frac{\text{surface area of the cylinder}}{\text{surface area of the sphere}}$$



25. Marion has mixed together ingredients for a cake. The recipe requires a baking tin that is cylindrical in shape with a diameter of 20 cm and a height of 5 cm. Marion only has a tin in the shape of a trapezoidal prism and a muffin tray consisting of 24 muffin cups. Each of the muffin cups in the tray is a portion of a cone. Both the tin and muffin cup are shown in the diagrams.
Explain whether Marion should use the tin or the muffin tray.



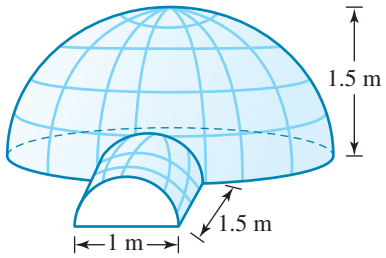


26. Sam is having his 16th birthday party and wants to make an ice trough to keep drinks cold. He has found a square piece of sheet metal with a side length of 2 metres.
He cuts squares of side length x metres from each corner, then bends the sides of the remaining sheet.
When four squares of the appropriate side length are cut from the corners, the capacity of the trough can be maximised at 588 litres.
Explain how Sam should proceed to maximise the capacity of the trough.



Problem solving

27. Nathaniel and Annie are going to the snow for survival camp. They plan to construct an igloo, consisting of an entrance and a hemispherical living section as shown.



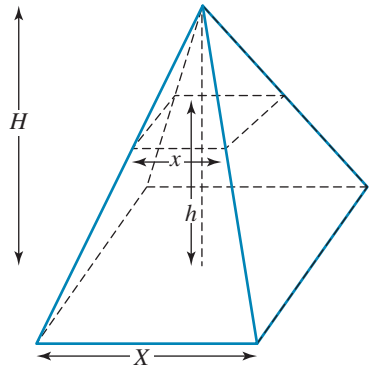
Nathaniel and Annie are asked to redraw their plans and increase the height of the liveable region (hemispherical structure) so that the total volume (including the entrance) is doubled.
Determine what must the new height of the hemisphere be to achieve this so that the total volume (including the entrance) is doubled. Write your answer in metres correct to 2 decimal places.

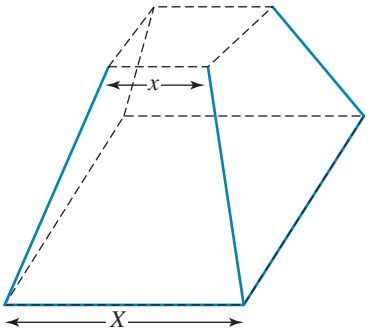
28. Six tennis balls are just contained in a cylinder as the balls touch the sides and the end sections of the cylinder. Each tennis ball has a radius of R cm.



a. Express the height of the cylinder in terms of R.
b. Evaluate the total volume of the tennis balls.
c. Determine the volume of the cylinder in terms of R.
d. Show that the ratio of the volume of the tennis balls to the volume of the cylinder is $2:3$.

29. A frustum of a square-based pyramid is a square pyramid with the top sliced off. H is the height of the full pyramid and h is the height of the frustum.





a. Determine the volume of the large pyramid that has a square base side of X cm.
b. Evaluate the volume of the small pyramid that has a square base side of x cm.
c. Show that the relationship between H and h is given by $H = \dfrac{Xh}{X-x}$.
d. Show that the volume of the frustum is given by $\dfrac{1}{3}h\left(X^2 + x^2 + Xx\right)$.

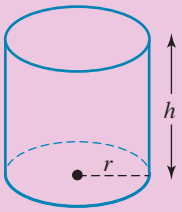
30. A large container is five-eighths full of ice-cream. After removing 27 identical scoops, it is one-quarter full. Determine how many scoops of ice-cream are left in the container.

LESSON
6.5 Review

6.5.1 Topic summary

Prisms and cylinders

- Prisms are 3D objects that have a uniform cross section and all flat surfaces.
- The surface area of a prism is calculated by adding the areas of its faces.
- The volume of a prism is $V = AH$, where A is the cross-sectional area of the prism, and H is the perpendicular height.
- A cylinder is a 3D object that has a circular cross-section.
- The curved surface area of a cylinder is $2\pi rh$, and the total surface area is $2\pi r^2 + 2\pi rh = 2\pi rh(r + h)$.
- The volume of a cylinder is $V = \pi r^2 h$.



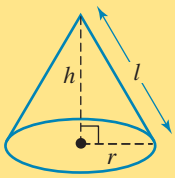
SURFACE AREA AND VOLUME

Pyramids

- The surface area of a pyramid can be calculated by adding the surface areas of its faces.
- The volume of a pyramid is $V = \frac{1}{3}AH$, where A is the area of the base and H is the height.

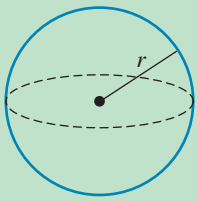
Cones

- The curved surface area of a cone is $SA_{curved} = \pi rl$, where l is the slant height.
- The total surface area is $SA = \pi rl + \pi r^2 = \pi r(l + r)$.
- The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.



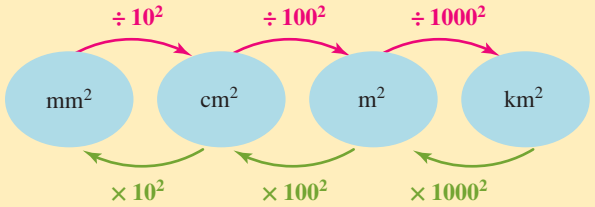
Spheres

- The surface area of a sphere is $A = 4\pi r^2$.
- The volume of a sphere is $V = \frac{4}{3}\pi r^3$.



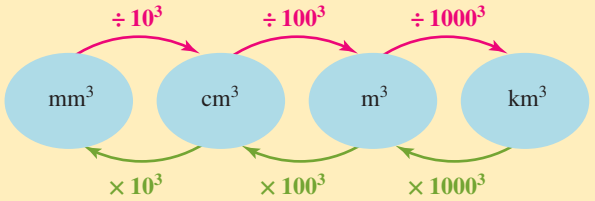
Units of area, volume and capacity

Area

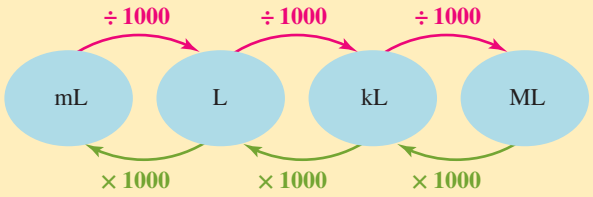


1 hectare (ha) = 10 000 m²

Volume



Capacity



1 cm³ = 1 mL
1 L = 1000 cm³

Area formulas

- Square: $A = l^2$
- Rectangle: $A = lw$
- Triangle: $A = \frac{1}{2}bh$
- Parallelogram: $A = bh$
- Trapezium: $A = \frac{1}{2}(a + b)h$
- Kite: $A = \frac{1}{2}xy$
- Circle: $A = \pi r^2$
- Sector: $A = \frac{\theta}{360} \times \pi r^2$
- Ellipse: $A = \pi ab$

Heron's formula

- Heron's formula is an alternate method to calculate the area of a triangle.
- For a triangle with sides of length a, b and c, the area is $A = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$.

6.5.2 Success criteria

Tick the column to indicate that you have completed the lesson and how well you have understood it using the traffic light system.

(**Green:** I understand; **Yellow:** I can do it with help; **Red:** I do not understand)

Lesson	Success criteria			
6.2	I can convert between units of area.			
	I can calculate the areas of plane figures using area formulas.			
	I can calculate the area of a triangle using Heron's formula.			
6.3	I can calculate the total surface areas of rectangular prisms and pyramids.			
	I can calculate the total surface areas of cylinders and spheres.			
	I can calculate the total surface areas of cones.			
6.4	I can calculate the volumes of prisms, including cylinders.			
	I can calculate the volumes of spheres.			
	I can calculate the volumes of pyramids.			

6.5.3 Project

So close!

Humans must measure! Imagine what a chaotic world it would be if we didn't measure anything. Some of the things we measure are time, length, weight and temperature; we also use other measures derived from these such as area, volume, speed.

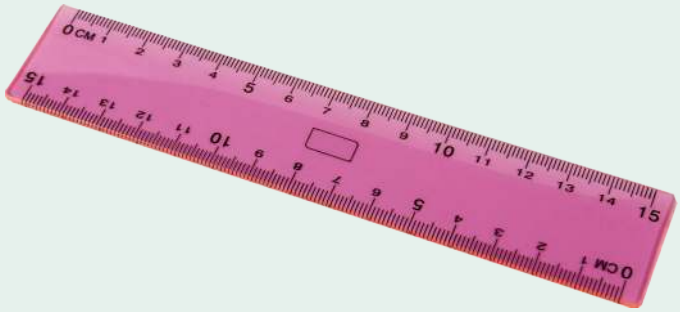
Accurate measurement is important. The accuracy of a measurement depends on the instrument being used to measure and the interpretation of the measurement. There is no such thing as a perfectly accurate measurement. The best we can do is learn how to make meaningful use of the numbers we read off our devices. It is also important to use appropriate units of measurement.



Measurement errors

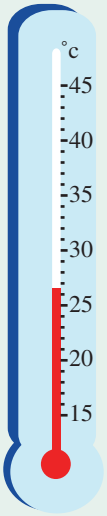
When we measure a quantity by using a scale, the accuracy of our measurement depends on the markings on the scale. For example, the ruler shown can measure both in centimetres and millimetres.

Measurements made with this ruler would have ± 0.5 mm added to the measurement. The quantity ± 0.5 mm is called the tolerance of measurement or measurement error.



$$\text{Tolerance of measurement} = \frac{1}{2} \times \text{size of smallest marked unit}$$

For a measurement of 5.6 ± 0.5 mm, the largest possible value is $5.6\text{ cm} + 0.5\text{ mm} = 5.65$ cm, and the smallest value is $5.6\text{ cm} - 0.5\text{ mm} = 5.55$ cm.



1. For the thermometer scale shown:
   - **a.** identify the temperature
   - **b.** state the measurement with its tolerance
   - **c.** calculate the largest and smallest possible values.
2. Calculate the largest and smallest values for:
   - **a.** $(56.2 \pm 0.1) - (19.07 \pm 0.05)$
   - **b.** $(78.4 \pm 0.25) \times (34 \pm 0.1)$.

Significant figures in measurement

A significant figure is any non zero-digit, any zero appearing between two non-zero digits, any trailing zeros in a number containing a decimal point, and any digits in the decimal places. For example, the number 345.6054 has 7 significant figures, whereas 300 has 1 significant figure.

The number of significant figures is an expression of the accuracy of a measurement. The greater the number of significant figures, the more accurate the measurement. For example, a fast food chain claims it has sold 6 000 000 000 hamburgers, not 6 453 456 102. The first measurement has only 1 significant figure and is a very rough approximation of the actual number sold, which has 10 significant figures.

Reducing the number of significant figures is a process that is similar to rounding.

Rounding and measurement error in calculations

When you perform calculations, it is important to keep as many significant digits as practical and to perform any rounding as the *final* step. For example, calculating 5.34×341 by rounding to 2 significant figures *before* multiplying gives $5.30 \times 340 = 1802$, compared with 1820 if the rounding is carried out after the multiplication.

Calculations that involve numbers from measurements containing errors can result in answers with even larger errors. The smaller the tolerances, the more accurate the answers will be.

3. a. Calculate $45\,943.4503 \times 86.765\,303$ by:

i. first rounding each number to 2 significant figures

ii. rounding only the answer to 2 significant figures.

b. Compare the two results.

Error in area and volume resulting from an error in a length measurement

The side length of a cube is measured and incorrectly recorded as 5 cm. The actual length is 6 cm. The effect of the length measurement error used on calculations of the surface area is shown below.

Error used in length measurement $= 1$ cm

Surface area calculated with incorrectly recorded value $= 5^2 \times 6 = 150\,\text{cm}^2$

Surface area calculated with actual value $= 6^2 \times 6 = 216\,\text{cm}^2$

$$\text{Percentage error} = \frac{216 - 150}{6} \times 100\% \approx 30.5\%$$

4. a. Complete a similar calculation for the volume of the cube using the incorrectly recorded length. What conclusion can you make regarding errors when the number of dimensions increase?

b. Give three examples of a practical situation where an error in measuring or recording would have a potentially disastrous impact.

Resources

eWorkbook Topic 6 workbook (worksheets, code puzzle and a project) (ewbk-13290)

Interactivities Crossword (int-2842)

Sudoku puzzle (int-3593)

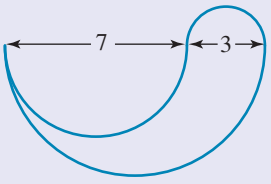
Exercise 6.5 Review questions

learnon

Unless told otherwise, where appropriate, give answers correct to 2 decimal places.

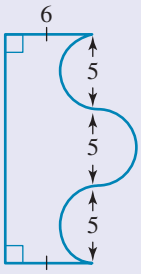
Fluency

1. MC If all measurements are in cm, the area of the figure is:



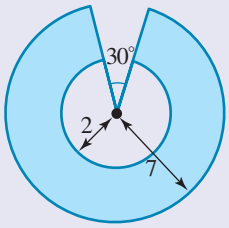
A. $16.49\,\text{cm}^2$ **B.** $39.25\,\text{cm}^2$ **C.** $9.81\,\text{cm}^2$ **D.** $23.56\,\text{cm}^2$

2. MC If all measurements are in centimetres, the area of the figure is:



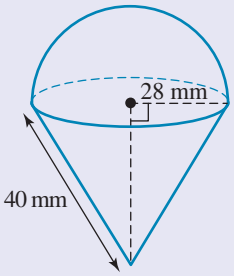
A. 50.73 cm^2
B. 99.82 cm^2
C. 80.18 cm^2
D. 90 cm^2

3. MC If all measurements are in centimetres, the shaded area of the figure is:



A. 3.93 cm^2 **B.** 11.52 cm^2 **C.** 388.77 cm^2 **D.** 129.59 cm^2

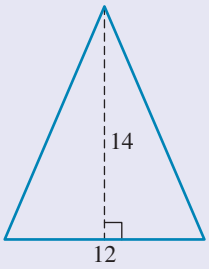
4. MC The total surface area of the solid is:



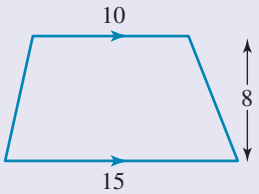
A. 8444.6 mm^2 **B.** 9221 mm^2 **C.** 14 146.5 mm^2 **D.** 50 271.1 mm^2

5. Calculate the areas of the following plane figures. All measurements are in cm.

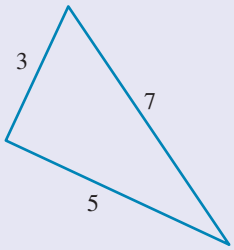
a.



b.

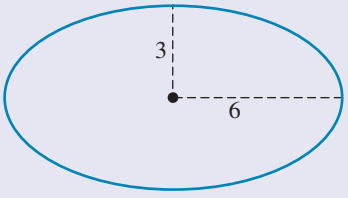


c.

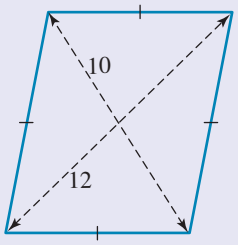


6. Calculate the areas of the following plane figures. All measurements are in cm.

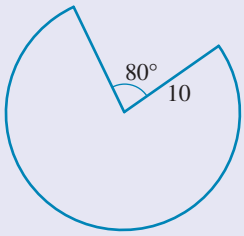
a.



b.

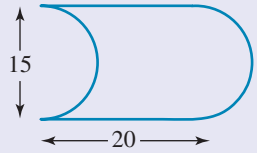


c.

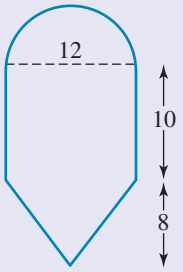


7. Calculate the areas of the following figures. All measurements are in cm.

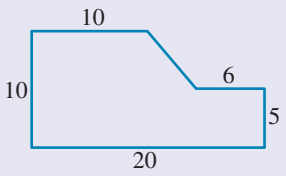
a.



b.

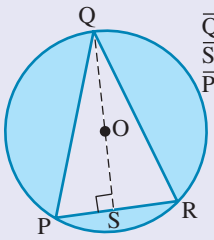


c.



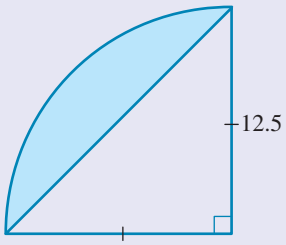
8. Calculate the blue shaded area in each of the following. All measurements are in cm.

a.

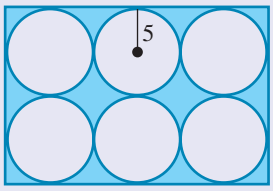


$\overline{QO} = 15$ cm
$\overline{SO} = 8$ cm
$\overline{PR} = 18$ cm

b.

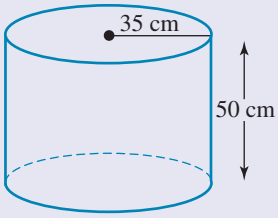


c.

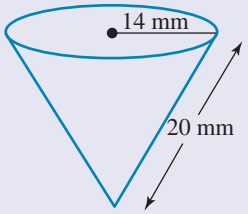


9. Calculate the total surface area of each of the following solids.

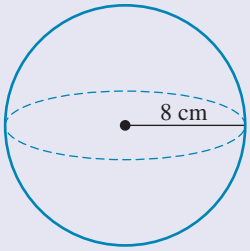
a.



b.

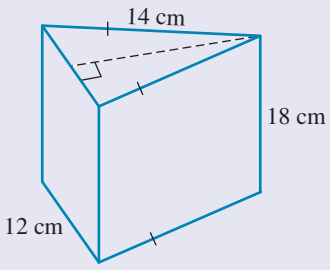


c.

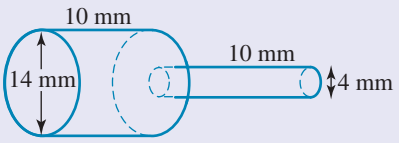


10. Calculate the total surface area of each of the following solids.

a.

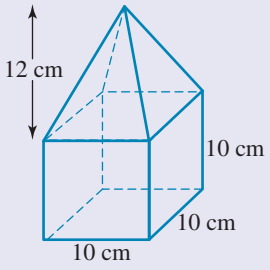


b.



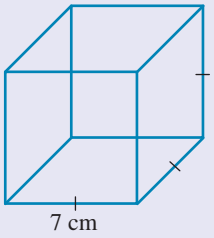
[closed at both ends]

c.

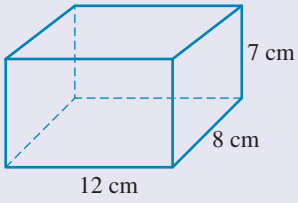


11. Calculate the volume of each of the following.

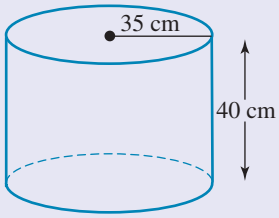
a.



b.

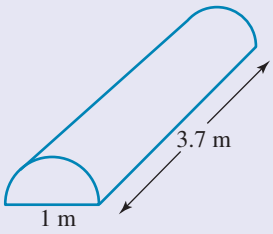


c.

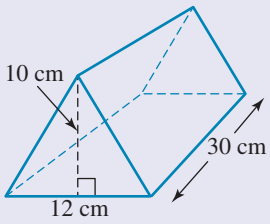


12. Determine the volume of each of the following.

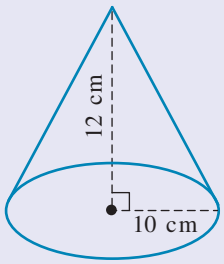
a.



b.

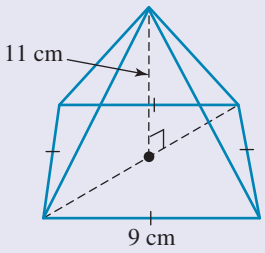


c.

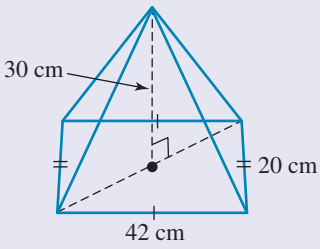


13. Determine the volume of each of the following.

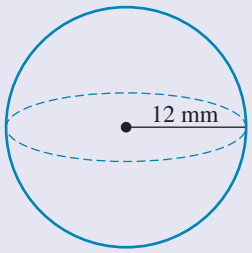
a.



b.



c.



Problem solving

14. A rectangular block of land 4 m × 25 m is surrounded by a concrete path 1 m wide.

a. Calculate the area of the path.

b. Determine the cost of concreting at $45 per square metre.

15. If the radius is tripled and the height of a cylinder is divided by six, then determine the effect on its volume (in comparison with the original shape).

16. If the length is halved, the width is tripled and the height of a rectangular prism is doubled, then determine the effect on its volume (in comparison with the original shape).

17. A cylinder of radius 14 cm and height 20 cm is joined to a hemisphere of radius 14 cm to form a bread holder.

a. Calculate the total surface area.

b. Determine the cost of chroming the bread holder on the outside at $0.05 per cm^2.

c. Calculate the storage volume of the bread holder.

d. Determine how much more space is in this new bread holder than the one it is replacing, which had a quarter circle end with a radius of 18 cm and a length of 35 cm.

18. Bella Silos has two rows of silos for storing wheat. Each row has 16 silos and all the silos are identical, with a cylindrical base (height of 5 m, diameter of 1.5 m) and conical top (diameter of 1.5 m, height of 1.1 m).



a. Calculate the slant height of the conical tops.

b. Determine the total surface area of all the silos.

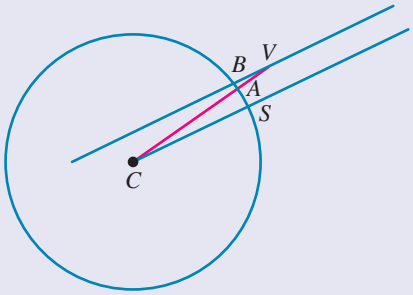
c. Evaluate the cost of painting the silos if one litre of paint covers 40 m^2 at a bulk order price of $28.95 per litre.

d. Determine how much wheat can be stored altogether in these silos.

e. Wheat is pumped from these silos into cartage trucks with rectangular containers 2.4 m wide, 5 m long and 2.5 m high.
Determine how many truckloads are necessary to empty all the silos.

f. If wheat is pumped out of the silos at 2.5 m^3/min, determine how long it will take to fill one truck.

19. The Greek mathematician Eratosthenes developed an accurate method for calculating the circumference of the Earth 2200 years ago!
The figure illustrates how he did this.



In this figure, A is the town of Alexandria and S is the town of Syene, exactly 787 km due south. When the sun's rays (blue lines) were vertical at Syene, they formed an angle of 7.2° at Alexandria ($\angle BVA = 7.2°$), obtained by placing a stick at A and measuring the angle formed by the sun's shadow with the stick.

a. Assuming that the sun's rays are parallel, evaluate the angle $\angle SCA$, correct to 1 decimal place.

b. Given that the arc $AS = 787$ km, determine the radius of the Earth, SC. Write your answer correct to the nearest kilometre.

c. Given that the true radius is 6380 km, determine Eratosthenes' percentage error, correct to 1 decimal place.



To test your understanding and knowledge of this topic, go to your learnON title at www.jacplus.com.au and complete the **post-test**.

Answers

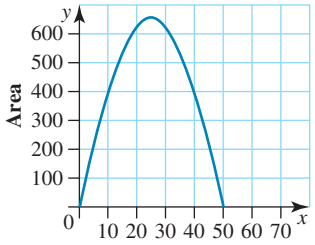
Topic 6 Surface area and volume

6.1 Pre-test

1. 68.63 mm^2
2. 125.7 cm^2
3. D
4. 706.9 cm^2
5. 4 cm^3
6. 57.7 cm^2
7. $864
8. B
9. 60 mm^3
10. C
11. D
12. 4.5 cm
13. C
14. 13.4 cm^2
15. 230 mL

6.2 Area

1. a. 16 cm^2 b. 48 cm^2 c. 75 cm^2
2. a. 120 cm^2 b. 706.86 cm^2 c. 73.5 mm^2
3. a. 254.47 cm^2 b. 21 m^2 c. 75 cm^2
4. a. 20.66 cm^2 b. 7.64 cm^2
5. a. 113.1 mm^2 b. 188.5 mm^2
6. a. i. 12π cm^2 ii. 37.70 cm^2
   b. i. $\frac{69\pi}{2}$ mm^2 ii. 108.38 mm^2
   c. i. 261π cm^2 ii. 819.96 cm^2
7. D
8. D
9. a. 123.29 cm^2 b. 1427.88 m^2 c. 52 cm^2
10. a. 30.4 m^2 b. 78 cm^2 c. 2015.50 cm^2
11. a. 125.66 cm^2 b. 102.87 m^2
12. a. 13.73 m^2 b. 153.59 m^2
13. a. 27.86 m^2 b. 37.5 m^2
14. 11 707.92 cm^2
15. 21 m^2
16. 60
17. $840
18. a. 260.87 m^2 b. 195.71 m^2 c. 75%
19. a. Sample responses can be found in the worked solutions in the online resources.
    b. 2020.83 m; horizontal. For a vertical split, 987.5 m.
20. a. $50 = x + y$
    b. $y = 50 - x$
    c. Area $= 50x - x^2$
    d. See the table at the bottom of the page.*
    e. No, impossible to make a rectangle.
    f.



    g. $x = 25$
    h. $y = 25$
    i. Square
    j. 625 m^2
    k. $r = 15.92$ m
    l. 795.77 m^2
    m. 170.77 m^2
21. a. Circular area, 1790.49 m^2; rectangular area, 1406.25 m^2
    b. Circular area, $\left(\frac{1}{4\pi}n^2\right)$ m^2; rectangular (square) area, $\left(\frac{1}{16}n^2\right)$ m^2. Circular area is always $\frac{4}{\pi}$ or 1.27 times larger.
22. a. 258.1 m^2 b. 7 bags c. $79
    d. $435.50
23. a. $\frac{29}{50}$ b. $x = 5, y = 5$
24. 32.88 cm^2

6.3 Total surface area

1. a. 600 cm^2 b. 384 cm^2 c. 1440 cm^2
   d. 27 m^2
2. a. 113.1 m^2 b. 6729.3 cm^2 c. 8.2 m^2
   d. 452.4 cm^2
3. a. 1495.4 cm^2 b. 502.7 cm^2
4. a. 506.0 cm^2 b. 9.4 m^2 c. 340.4 cm^2
   d. 224.1 cm^2
5. a. 13.5 m^2 b. 90 m^2 c. 11 309.7 cm^2
6. a. 9852.0 mm^2 b. 125.7 cm^2 c. 1531.4 cm^2
7. a. 880 cm^2 b. 3072.8 cm^2 c. 75 cm^2
8. a. 70.4 cm^2 b. 193.5 cm^2 c. 1547.2 cm^2
9. B
10. a. 70.0 m^2 b. $455
11. a. 3063.1 cm^2 b. $168.47
12. 11 216 cm^2

*20. d.

x	0	5	10	15	20	25	30	35	40	45	50
Area(m^2)	0	225	400	525	600	625	600	525	400	225	0

13. 60

14. 1592 cm^2

15. $1960

16. **a.** 8π m^2 **b.** $\frac{\sqrt{7}}{2}$ m **c.** $4\sqrt{2}\pi$ m^2

17. **a.** 6.6 m^2

b. Back wall = 80 tiles
Side wall = 50 tiles
80 + 50 + 50 = 180 tiles

c. Cheapest: 30 cm by 30 cm, $269.50; 20 cm by 20 cm (individually) $270; 20 cm by 20 cm (boxed) $276.50

18. The calculation is correct.

19. **a.** $\theta = 120°$ **b.** $x = 1$; $y = \sqrt{3}$
c. $3\sqrt{3}$ cm^2 **d.** $6\sqrt{3}$ cm^2
e. 32

20. The area of material required is 1.04 m^2. If Tina is careful in placing the pattern pieces, she may be able to cover the footstool.

21. $r = \frac{3\sqrt{3}a}{2}$

22. **a.** Arc length XY $= (x + s)\theta$
Arc length AB $= x\theta$

b. i. $x = \frac{2\pi t}{\theta} = \frac{st}{r - t}$

ii. $\frac{x}{x + s} = \frac{t}{r}$

c. Area of sector AVB $= \frac{x^2\theta}{2}$

Area of sector XVY $= \frac{(s + x)^2\theta}{2}$

Area of ABYX $= \frac{s\theta\,(s + 2x)}{2}$

TSA of frustum $= \pi\left(t^2 + r^2\right) + \frac{s\theta\,(s + 2x)}{2}$

6.4 Volume

1. **a.** 27 cm^3 **b.** 74.088 m^3 **c.** 3600 cm^3
d. 94.5 cm^3

2. **a.** 450 mm^3 **b.** 360 cm^2

3. **a.** 6333.5 cm^3 **b.** 19.1 m^3 **c.** 280 cm^3

4. **a.** 288 mm^3 **b.** 91.6 m^3 **c.** 21 470.8 cm^3

5. **a.** 7.2 m^3 **b.** 14 137.2 cm^3 **c.** 1436.8 mm^3
d. 523 598.8 cm^3

6. **a.** 113 097.34 cm^3 **b.** 1.44 m^3
c. 12 214.51 mm^3 **d.** 101.93

7. **a.** 377.0 cm^3 **b.** 2303.8 mm^3

8. **a.** 400 cm^3 **b.** 10 080 cm^3

9. **a.** 1400 cm^3 **b.** 10 379.20 cm^3

10. **a.** 41.31 cm^3 **b.** 48.17 cm^3

11. **a.** 218.08 cm^3 **b.** 3691.37 cm^3

12. **a.** $V_{new} = 27l^3$; the volume will be 27 times as large as the original volume.

b. $V_{new} = \frac{1}{8}l^2$; the volume will be $\frac{1}{8}$ of the original volume.

c. $V_{new} = 2\pi r^2 h$; the volume will be twice as large as the original volume.

d. $V_{new} = \pi r^2 h$; the volume will remain the same.

e. $V_{new} = 3lwh$; the volume will be 3 times as large as the original value.

13. D

14. **a.** 256 cm^3 **b.** 5120 cm^3 **c.** 12 000 cm^3
d. 6880 cm^3 **e.** 7438.35 cm^3 **f.** 558.35 cm^3

15. 4417.9 L

16. 10 215.05 cm^3

17. **a.** $H = 6$ m **b.** 112 m^3 **c.** 19 bins
d. 112 000 L **e.** 1.95 m from floor

18. **a. i.** 4.57 cm **ii.** 262.5 cm^2

b. i. 14.15 cm **ii.** 323.27 cm^2

c. i. 33.33 cm **ii.** 437.62 cm^2

d. Sphere — costs less for a smaller surface area.

19. **a.** $h = \frac{V}{\pi r^2}$

b. i. 31.8 cm **ii.** 8.0 cm

c. $\sqrt{\frac{V}{\pi h}}$

d. $r \geq 0$, since r is a length.

e. i. 7.6 cm **ii.** 6.2 cm

20. 1000

21. **a.** 2.55 cm

b. 35.68 cm

c. $A_a = 157.88$ m^2, $A_b = 12.01$ m^2

22. **a.** 126.67 m^3 **b.** 53.33 m^3

23. Volume of water needed: 30.9 m^3

24. Sample responses can be found in the worked solutions in the online resources.

25. Required volume = 1570.80 cm^3; tin volume = 1500 cm^3; muffin tray volume = 2814.72 cm^3. Marion could fill the tin and have a small amount of mixture left over, or she could almost fill 14 of the muffin cups and leave the remaining cups empty.

26. Cut squares of side length $s = 0.3$ m or 0.368 m from the corners.

27. 1.94 m

28. **a.** $H = 12R$ **b.** $8\pi R^3$ **c.** $12\pi R^3$
d. 8 : 12 = 2 : 3

29. **a.** $\frac{1}{3}X^2H$ **b.** $\frac{1}{3}x^2(H - h)$

c, d. Sample responses can be found in the worked solutions in the online resources.

30. 18 scoops

Project

1. a. The temperature reading is 26.5 °C.
   b. The smallest unit mark is 1°C, so the tolerance is 0.5.
   c. Largest possible value = 27 °C,
   smallest possible value = 26 °C
2. a. Largest value = 37.28, smallest value = 36.98
   b. Largest value = 2681.965, smallest value = 2649.285
3. a. i. 4 002 000
   ii. 4 000 000
   b. The result for i has 4 significant figures, whereas ii has only 1 significant figure after rounding. However, ii is closer to the actual value (3 986 297.386 144 940 9).
4. Volume using the incorrectly recorded value = 125 cm^3
   Volume using the actual value = 216 cm^3
   The percentage error is 42.1%, which shows that the error compounds as the number of dimensions increases.

6.5 Review questions

1. D
2. C
3. D
4. A
5. a. 84 cm^2 b. 100 cm^2 c. 6.50 cm^2
6. a. 56.52 cm^2 (when $\pi = 3.14$) or 56.55 cm^2 (when $\pi = 3.1416$)
   b. 60 cm^2
   c. 244.35 cm^2
7. a. 300 cm^2 b. 224.55 cm^2 c. 160 cm^2
8. a. 499.86 cm^2 b. 44.59 cm^2 c. 128.76 cm^2
9. a. 18 692.48 cm^2 b. 1495.40 mm^2 c. 804.25 cm^2
10. a. 871.79 cm^2 b. 873.36 mm^2 c. 760 cm^2
11. a. 343 cm^3 b. 672 cm^3 c. 153 938.04 cm^3
12. a. 1.45 m^3 b. 1800 cm^3 c. 1256.64 cm^3
13. a. 297 cm^3 b. 8400 cm^3 c. 7238.23 mm^3
14. a. 62 m^2 b. $2790
15. $V = \frac{3}{2}\pi r^2 h$; the volume will be 1.5 times as large as the original volume.
16. $V = 3lwh$; the volume will be 3 times as large as (or triple) the original volume.
17. a. 3606.55 cm^2 b. $180.33 c. 18 062.06 cm^3
    d. 9155.65 cm^3
18. a. 1.33 m
    b. 910.91 m^2
    c. $618.35 or $636.90 assuming you have to buy full litres (i.e not 0.7 of a litre)
    d. 303.48 m^3
    e. 11 trucks
    f. 12 minutes
19. a. 7.2° b. 6263 km c. 1.8% error